EBN

122

EBN 122

18-07-2017

Chapter 1:

1. Units:

Mega	10^{6}	M
kilo	10^{3}	k
milli	10^{-3}	m
micro	10^{-6}	μ
Nano	10^{-9}	n
Pico	10^{-12}	p

Basic Concepts:

1.2. Electric Charge: (c) → electric prop. of particles.

$q(e^-) = -1.602 \times 10^{-19} C$

$1\ C = \frac{1}{e} = 6.24 \times 10^{18}\ e^-$

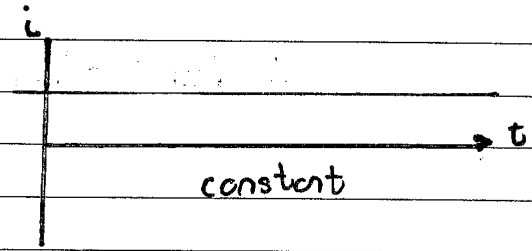
1.3. Current: (A)

$$i \triangleq \frac{dq}{dt} = \frac{\Delta q}{\Delta t}$$

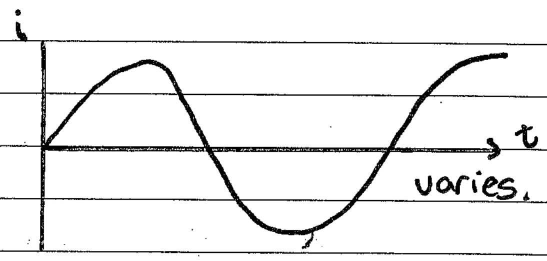
*Can be the slope of a graph.

(charge) $Q = \int_{t_0}^{t} i\, dt.$

DC current:

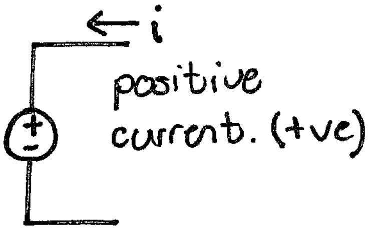


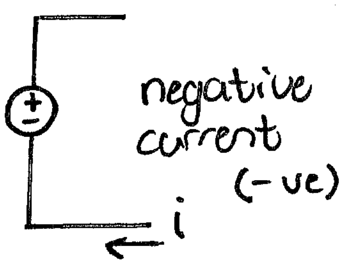
AC current:



$\therefore$ With DC : $Q = i \times \Delta t.$

With AC : $Q = \int_{t_0}^{t} i\, dt.$





* Across a circuit or over an component.

1.4 Voltage: $\longrightarrow$ energy required to move 1 unit of charge. (V)

- work is done by emf.

w = energy (J)
q = charge (C)

- $V_{ab} = \frac{dw}{dq} = \frac{\Delta w}{\Delta q}$

- $V_{ab} = V_a - V_b$.

* $V_{ab} > 0$ potential of a ↑ potential of b.

* $V_{ab} < 0$ potential of a ↓ potential of b.

$\therefore V_{ab} = -V_{ba}$

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18-07-2017.

Note: protons +
∴ current
charge +

Examples:

① Calc. ammount of charge in 6mil protons.

$Q = (6\times10^{6})(1.602\times10^{-19})$
$= 9.612\times10^{-13}\ C$

② A rechargable battery delivers 60mA for 18h. How much charge can be delivered?

$Q = \int_{t_0}^{t_i} I\,dt = I\Delta t.$

$t = 18\times66\times60 = 64800s$

$Q = (60\times10^{-3}\ A)(64800\ s)$
$= 3888C$

③ Given the total charge of a terminal
$q = 10 - 10e^{-2t}\ mC$
find the current @ 1.0s.

$I = \frac{dq}{dt}$ $\quad I = \frac{d}{dt}(10-10e^{-2t})$
$= 20\ e^{-2t}\ mA$

$I_{(1)} = 20e^{-2(1)}\ mA$
$= 2.707\ mA.$

$q = i\times t.$

④ The current through an element is represented by

$$i = \begin{cases} 4\ A & 0 < t < 1 \\ 4t^2\ A & t > 1 \end{cases} \quad \text{from } 0 \to 2$$

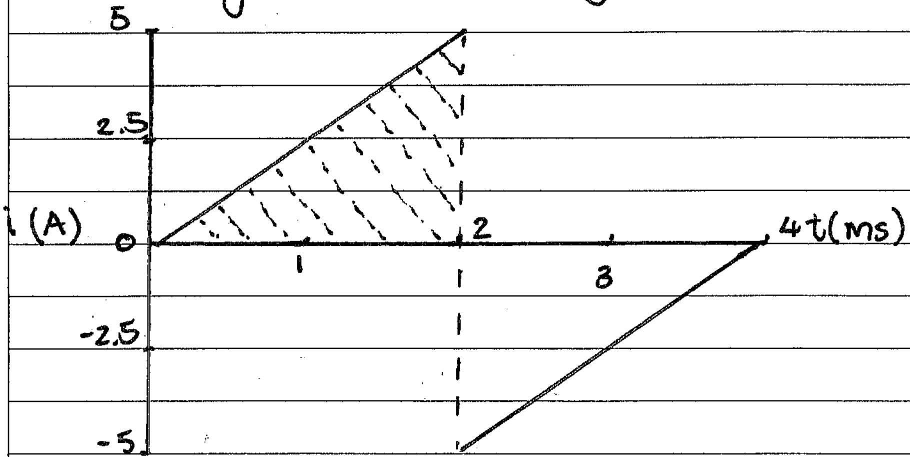
$$Q = \int_{t_0}^{t_2} I\, dt.$$

$$Q = \int_0^1 4A\,dt + \int_1^2 4t^2\ A\ dt$$

$$= 4 + \left(\frac{4}{3}(2)^3 - \frac{4}{3}(1)^3 \right)$$

$$= 13.333\ C$$

⑤ The change in i through an element



$$Q = \int_0^2 i\ dt$$

$$= \frac{1}{2}(2)(5)$$

$$= \underline{5\ mC}$$

EBN 122

19-07-2017

Unit 1.5

Power & Energy

→ Rate at which (w) energy is consumed / delivered.

* See where current is entering.

$$p = \frac{dw}{dt} = \frac{dw}{dq} \times \frac{dq}{dt} = v \cdot i$$

(a) DC system: $p(t) = P = VI$ - constant.

(b) AC system: $p(t) = v(t)\, i(t)$ - time function.

$\sum p = 0$

$\Rightarrow p_{delivered} = p_{absorbed}$ } Concervation of energy.

Energy: → Capacity to do work.

$1\,Wh = 3600\,J$

$1\,kWh = 3.6\,MJ$

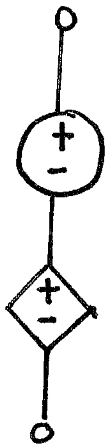
- $w = \int_{t_0}^{t} p \, dt$ Note: $p = v \cdot i$
- $w = \int_{t_0}^{t} v \cdot i \, dt$.

Unit 1.6.

Circuit Elements:

1. Active elements (Energy sources)

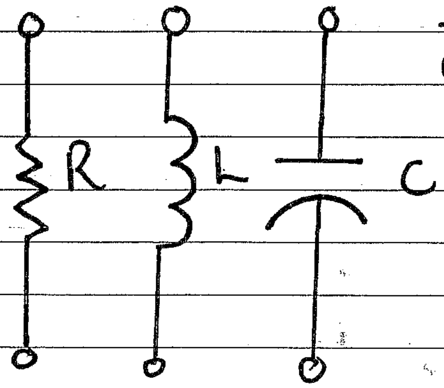
2 Types:



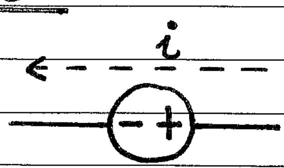
(a) Independent Source:
- Round

(b) Dependant Source:
CCVS, VCCS, CCCS, VCVS
- Square.

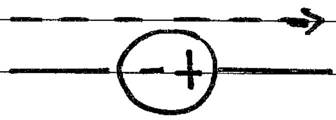
2. Passive elements: Resistors (R)
Inductors (L)
Capacitors (C)



Note:

1.


Current from + to −
then
$$p = +v \cdot i$$

2.


Current from − to +
then
$$p = -v \cdot i$$

if $p < 0$ then supplied.
$p > 0$ then absorbed.

EBN 122

19-07-2017.

1.5 & 1.6 Examples:

① To move a charge $q = 6C$ from $a \rightarrow b$ requires $-30J$: Find the Voltage drop. V_{ab}. $q = 6C$ $w = -30J$

$$V_{ab} = \frac{dw}{dq} = \frac{\Delta w}{\Delta q}$$
$$= \frac{-30}{6} = -5V$$

② Find the power delivered to on element at $t = 5ms$ if current entering is
$i = 5\cos 60\pi t$ A &
$v = 10 + 5\int_{t_0}^{t} i\, dt$ V

$$\Rightarrow V = 10 + 5\int_{t_0}^{t} 5\cos 60\pi t\, dt$$
$$= 10 + \frac{5}{12\pi}(\sin 60\pi t)$$

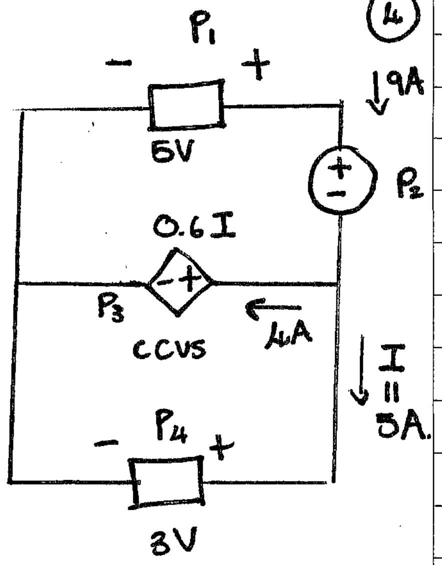
* $p = +v \cdot i$

$$p = \{5\cos 60\pi(5\times10^{-3})\}\{10 + \frac{5}{12\pi}(\sin 60\pi(5\times10^{-3}))\}$$
$$p = 29.7W$$

③ An element draws 15A when connected to a 240V line. How long wil it take to consume 180 kJ.

$V = 240V$
$i = 15A$
$w = 180\,000\,J$

$$w = \int_{t_0}^{t} p\, dt = p\Delta t$$
$$\Delta t = \frac{w}{p} = \frac{180\,000}{240 \times 15}$$
$$\Delta t = 50s$$



④ Calculate the power absorbed in each component.

$p_1: P = -V \cdot i$
$= -45W$ (power supply)

$p_2: P = V \cdot i$
$= 18W$

$p_3: V = 0.6(5) = 3V$
$P = 12W$

$p_4: P = V \cdot i$
$= 15W.$

⑤ A battery supplies a constant 1mA (DC) at 5V for 1 day. Determine the energy used.

$t = 86400s$
$V = 5V$
$i = 1mA$

$p = V \cdot i$
$= 5 \times (1 \times 10^{-3})$
$= 5 \times 10^{-3}$ W

★ $W = p\Delta t$
$W = (5 \times 10^{-3})(86400)$
$= 432$ J.

Elec. cost: R5/kWh
$p = 3kW$

⑥ A 3kW kettle needs 20s to boil 100mℓ of water. What is the cost of 1.5ℓ of water to boil?

Cost of 100 mℓ : $3 \times 5 \times (20 \div 60 \div 60)$
$= R0,0833$

Cost of 1.5ℓ = $\frac{0,0833}{100} \times 1500$
$= R\ 1.25.$

EBN 112 Chapter 2: 24-07-2017.

Chapter 2: Basic Laws:

Ohm's Law: (gradient of graph = R) $\{V = I \times R\}$

$$\text{gradient} = \frac{\Delta V}{\Delta i}$$

$$\Rightarrow R = \frac{V}{I}$$

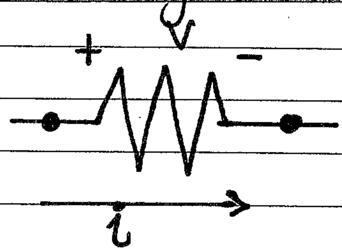
$$\frac{1}{R} = \frac{I}{V} = G \text{ (conductance)}$$

* The voltage across a resistor is directly prop. to the current I.

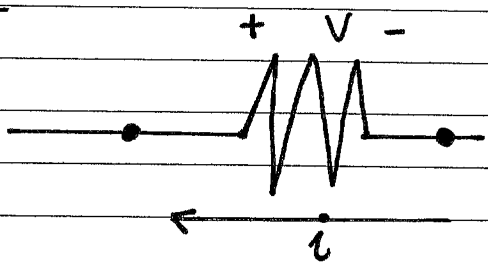
$$R = \rho \frac{\text{Length.}}{\text{Area}} = \Omega$$

ρ = resistivity.

Passive Sign Convention:



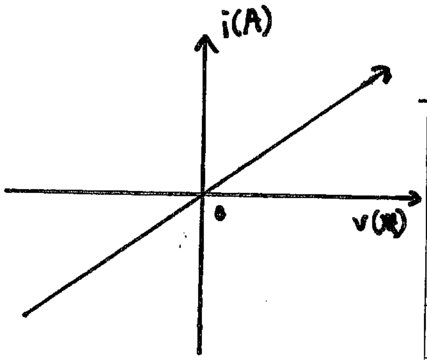
$V = +iR$

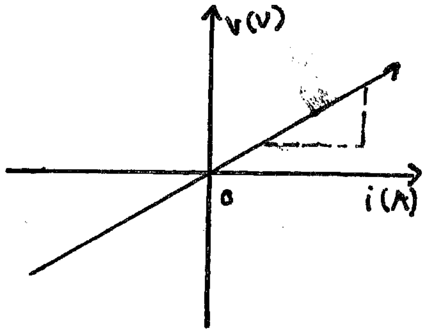


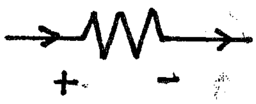
$V = -iR$

Resistance Extremes:

R = 0 -> Short Circuit   R = ∞ -> Open circuit.

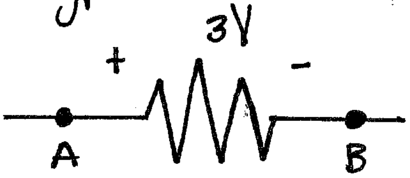






R depends on:

- Length
- Size
- Type.



$V_{ab} = 3V$

$V_{ba} = -3V.$

Conductance:

$G = \frac{1}{R}$ [S] $\qquad$ R [Ω]

$i = G \cdot V$

(2.2.) **Power dissipated in a Resistor:**

① $P = V \cdot I$ $\qquad$ ② $V = IR$

Power: * $p = \frac{V^2}{R}$ $\qquad$ $p = I^2 \times R.$

$\therefore \underline{\underline{R \geq 0}}$

EBN 112.

24-07-2017.

(2.3.)

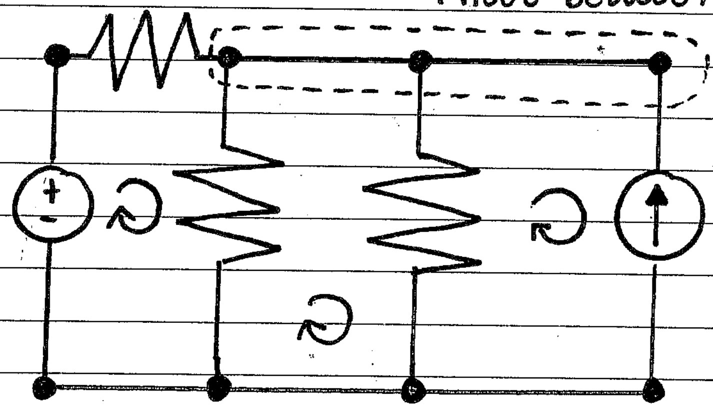
Nodes, Branches & Loops

① **Branch**: A single element. (5 Branches)
(2 terminals.

② **Node**: Point of connection between 2 or more branches. (3 Nodes)

③ **Loops**: Any closed path in a circuit.
(3 Loops).
* Can't cross an element twice.

1 node between 3 elements.



Independent Loops:

A loop that contains at least 1 branch that is not part of any other independent loops.

$$\boxed{b = l + n - 1}$$

Branches

loops

nodes.

Series & Parallel:

EBN 122

Chapter 2.

25-07-2017

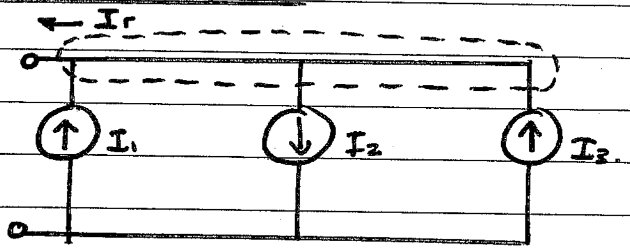
Kirchoff's Law:

Current Law: KCL:

→ Algebraic sum of currents entering & exiting a node is equal to 0. (or closed boundry).

$$\sum_{n=1}^{n} i_n = 0.$$

Current in Parallel:



$-I_r + I_1 - I_2 + I_3 = 0.$

∴ Current sources can NOT be connected in series.

Voltage law: KVL:

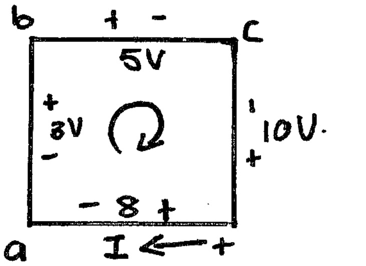
→ Algebraic sum of voltages (analysed clockwise) will be = 0 (around a closed path).

∴ Voltage sources can NOT be connected in parallel.

$KVL = \sum V_n = 0.$

Note: create a point on the node • & analyse current from that point.

- (enter a node + exit a node −)



$-3+5-10+8 = 0$

- Enter − then −V, Enter + then +V.

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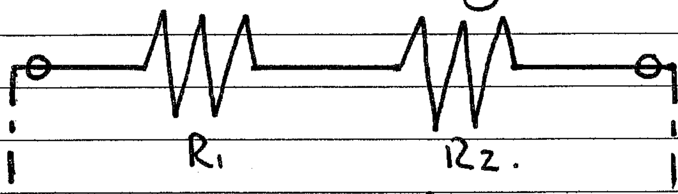
Series Resistors & Voltage Division

Series:

- Two or more elements are in series if they are cascaded & carry the same current (I)



$$V = i(R_1 + R_2)$$

$R_{equivalent} = R_1 + R_2$.

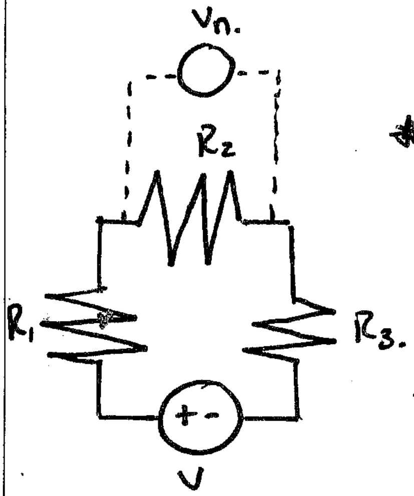
(2.5) **Voltage Division:** (resistors in series).

$R = \frac{V}{I}$

$$V = i R_{eq} \Rightarrow i = \frac{V}{R_1 + R_2}$$

$$\Rightarrow V_1 = \frac{V}{R_1 + R_2} \times R_1$$

$$V_2 = \frac{V}{R_1 + R_2} \times R_2$$



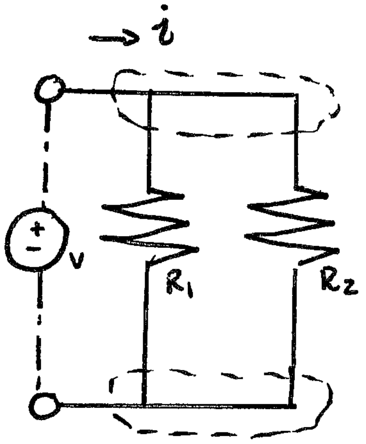
* **Req for more than 2 R's:**

$$R_{eq} = \sum_{n=1}^{n} R_n$$

$$\Rightarrow V_n = \frac{R_n}{\sum_{n=1}^{n} R_n} \times V.$$

voltage divided proportionally.

(2.6) Parallel Resistors & Current Division.



Parallel:

- Two ore more elements are in ∥ if they connect to the same 2 nodes (Same Voltage across them).

$$i = \frac{V}{R_{eq}} \qquad i = i_1 + i_2.$$

$$i = V \times G_{eq} \qquad \text{OR} \qquad i = \frac{V}{R_1} + \frac{V}{R_2}$$

$$G_{eq} = G_1 + G_2$$

$$G_{eq} = \frac{1}{R_1} + \frac{1}{R_2}. \qquad i = V\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

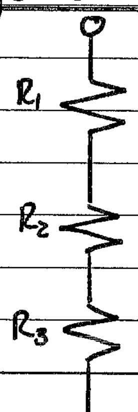
*. $$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_2 + R_1}{R_2 \times R_1}.$$

(2.6) Current Division: (resistors in parallel).

$$i_n = \frac{G_n}{G_1 + G_2 + \cdots G_n}\, i$$

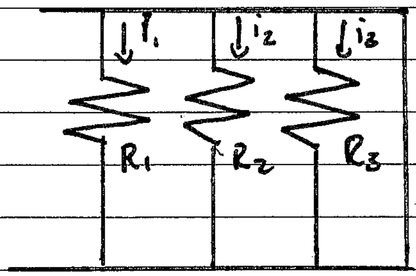
$$G_{eq} = \sum_{n=1}^{n} G_n.$$

Voltage Division summary:



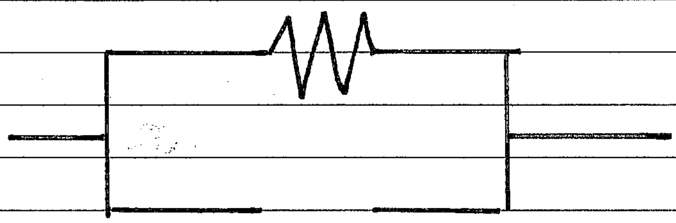
$$V_2 = \frac{R_2 \quad V}{R_1 + R_2 + R_3}$$

Current Division:



$$i_2 = \frac{G_2 \, i}{G_1 + G_2 + G_3}.$$

Adding Resistors:



$$R_1 \| R_\infty = \left(\frac{1}{R_1} + \frac{1}{R_\infty}\right)^{-1}$$

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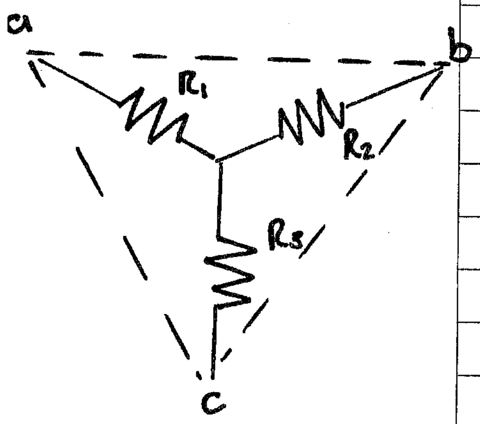
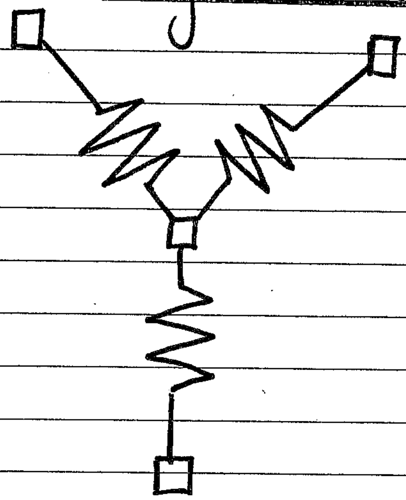
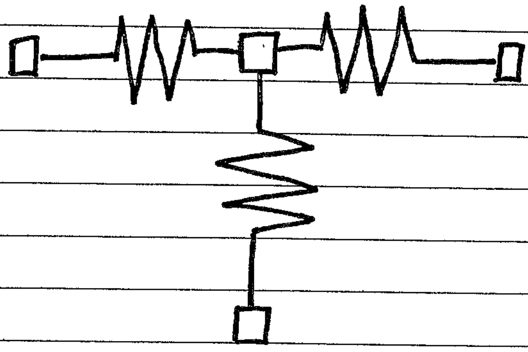
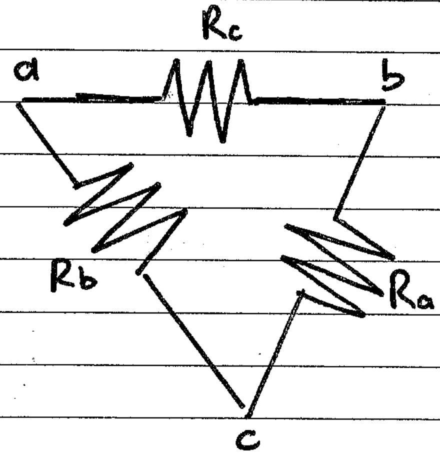


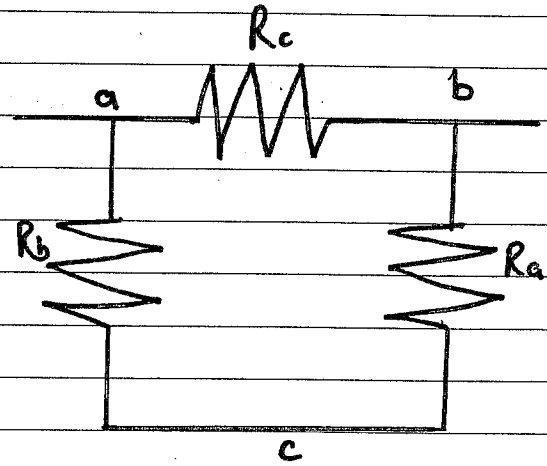
Wye-*-T Network:



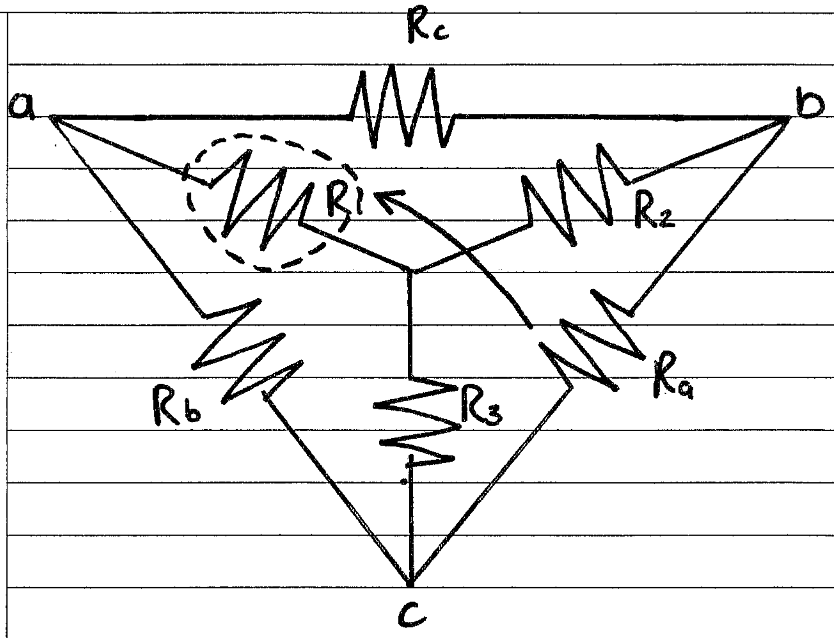


Delta or Pi network:





Why? Sometimes resistors are not in || or series.



adjacent Δ-resistors.

$$R_1 = \frac{R_b \times R_c}{R_a + R_b + R_c}$$ ← Σ of Δ resistors.

$$R_2 = \frac{R_c \times R_a}{R_a + R_b + R_c}$$

$$R_a = \frac{R_1R_2 + R_2R_3 + R_1R_3}{R_1}$$ * All multiples possible

opposite Y-resistor.

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Chapter 3:

3 Laws:

- → KCL : Nodal Anal.
- → KVL : Mesh.
- → Ohm's Law.

Nodal Analysis:

KCL: Nodal Analysis.
(Voltage Node analysis)

KVL: Mesh Analysis.
(Current analysis)

Steps: (For analysing)
(Node Voltages).

1. **Select:** A ref. (ground) node.
2. **Assign:** Voltages to remaining "critical" nodes i.e. $V_1, V_2, V_3, V_4 \cdots$
3. **Apply KCL:** To non-reference nodes. (Ohm's Law).
4. **Solve:**

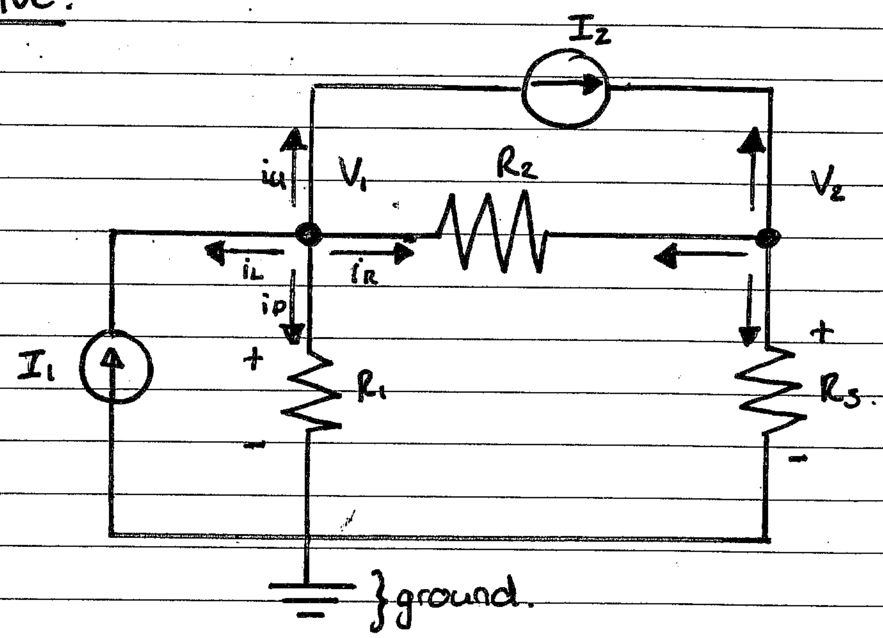
- Critical Node:
  - * Node with more than 2 branches.

- Current OUT of node (unless CCCS)
- Current flows from High to Low volt.

$\therefore i = \dfrac{V_{High} - V_{Low}}{R_{Branch}}$

$\sum i_{(node)} = 0.$

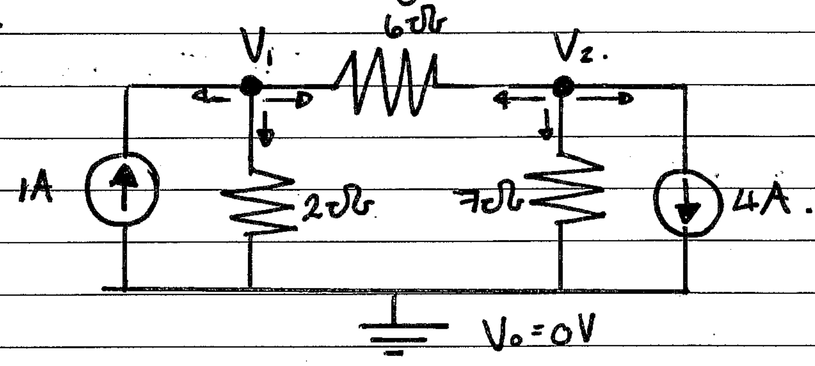
e.g. 1.



Ref. Node: @ - terminal of Voltage source.

Analyse Node.

e.g.2.



Method 1:

- Elimination:

For V_1: $+1A - \frac{V_1 - V_2}{6} - \frac{V_1 - 0}{2} = 0. \Big\} \times -6$

$-6 + V_1 - V_2 + 3V_1 = 0$

① $4V_1 - V_2 = 6.$

For V_2: $-4A - \frac{V_2 - V_0}{7} - \frac{V_2 - V_1}{6} = 0 \Big\} \times -42.$

$168 + 6V_2 + 7V_2 - 7V_1 = 0$

② $13V_2 - 7V_1 = -168$

Solve: $V_2 = -14V$
$V_1 = -2V.$

Method 2:

- Cramer's Rule:

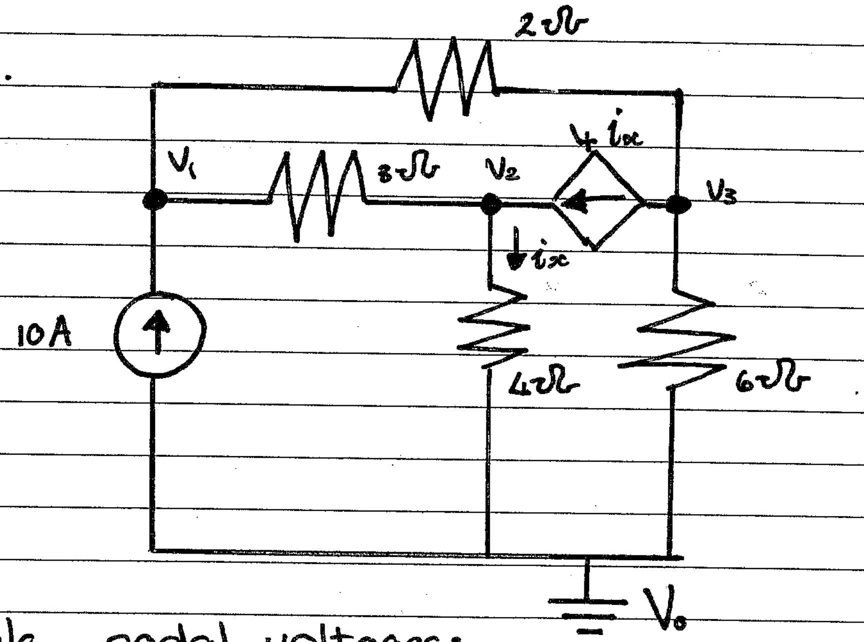
(Find the Determinant) {WTW}

① $4V_1 - 1V_2 = 6$

② $13V_2 - 7V_1 = -168.$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{\begin{vmatrix} 6 & -1 \\ -168 & 13 \end{vmatrix}}{\begin{vmatrix} 4 & -1 \\ -7 & 13 \end{vmatrix}} = \frac{-90}{45} = -2V.$$

e.g. 3.



1. Calc. nodal voltages:

@ V_1: $10A - \frac{V_1 - V_2}{3} - \frac{V_1 - V_3}{2} = 0 \Big\} \times -6$

$-60A + 2V_1 - 2V_2 + 3V_1 - 3V_3 = 0$

(1) $5V_1 - 2V_2 - 3V_3 = 60$

@ V_2: $\frac{V_2 - V_0}{4} + \frac{V_2 - V_1}{3} - 4i_x = 0 \Big\} \times 12.$

$3V_2 + 4V_2 - 4V_1 - 12V_2 = 0.$

(2) $-4V_1 - 5V_2 + 0V_3 = 0.$

* $i_x = \frac{V_2 - V_0}{4}$

* $i_x = \frac{V_2}{4}$

@ V_3: $\frac{V_3 - V_1}{2} + \frac{V_3 - V_0}{6} + 4i_x = 0 \Big\} \times 6$

(3) $-3V_1 + 6V_2 + 4V_3 = 0.$

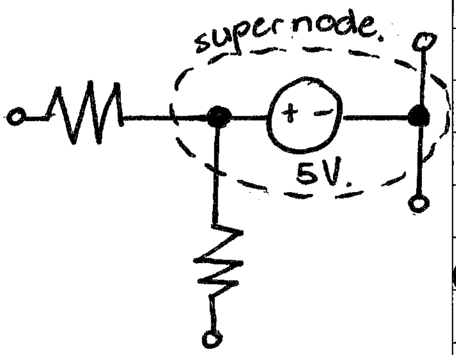
Matrix: $\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}$

$V_1 = \frac{\begin{vmatrix} 60 & -2 & -3 \\ 0 & -5 & 0 \\ 0 & 6 & 4 \end{vmatrix}}{\begin{vmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{vmatrix}}$ $\qquad V_2 = \frac{\begin{vmatrix} 5 & 60 & -3 \\ -4 & 0 & 0 \\ -3 & 0 & 4 \end{vmatrix}}{\begin{vmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{vmatrix}}$

3.3.

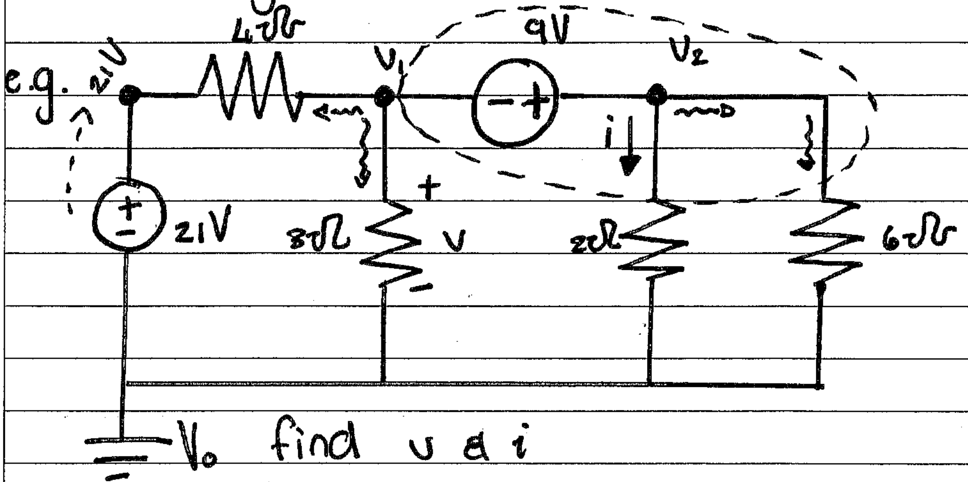
Super Node:

Voltage source enclosed by non-ref. node as well as elements connected in ∥ with the voltage source.



* 1^{st} find ΔV over super node. (1^{st} equation).

e.g.



find v & i

① $V_2 - V_1 = 9$ KCL @ supernode:

$$\left.\frac{V_1-21}{4} + \frac{V_1}{3} + \frac{V_2}{2} + \frac{V_2}{6} = 0\right\} \times 12$$

② $7V_1 + 8V_2 = 63$

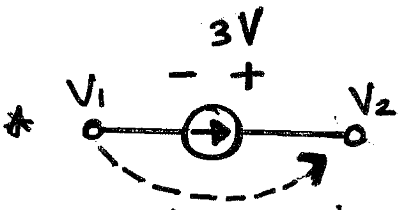
Cramer's Rule:

$$\begin{bmatrix} -1 & 1 \\ 7 & 8 \end{bmatrix}\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 9 \\ 63 \end{bmatrix}$$

$$V_1 = \frac{\begin{vmatrix} 9 & 1 \\ 63 & 8 \end{vmatrix}}{\begin{vmatrix} -1 & 1 \\ 7 & 8 \end{vmatrix}} \quad \frac{9}{-15} \qquad V_2 = \frac{\begin{vmatrix} -1 & 9 \\ 7 & 63 \end{vmatrix}}{\begin{vmatrix} -1 & 1 \\ 7 & 8 \end{vmatrix}} = \frac{-126}{-15}$$

$V_1 = -0.6\,V$ $V_2 = 8.4\,V.$

*



voltage gain from − to +.

$\Rightarrow V_1 + 3V = V_2$

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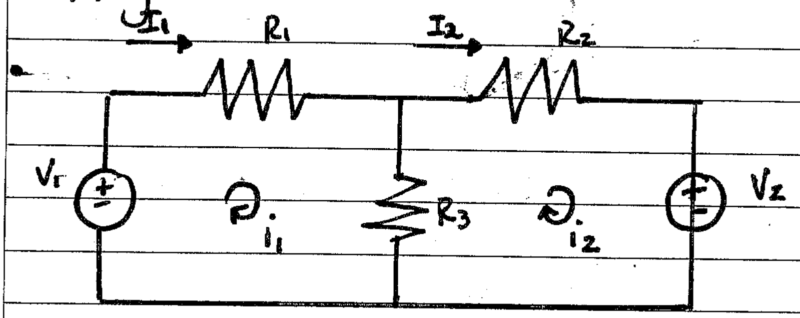
Mesh Analysis:

- A mesh is a loop that does not include any other loop.

- Apply KVL to find unknown current.



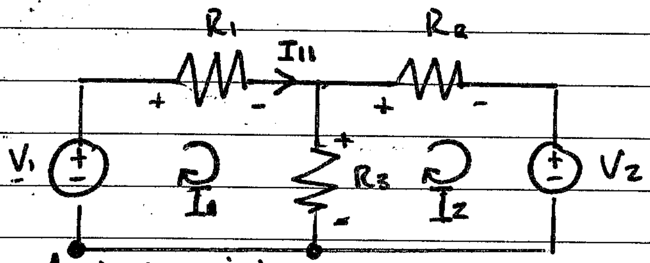
* 2 Meshes.
* 3 Loops.

Step 1: Assign Mesh current. → Label. → Direction.

Step 2: KVL → equations.

Step 3: Solve for the unknown.

e.g. 1.



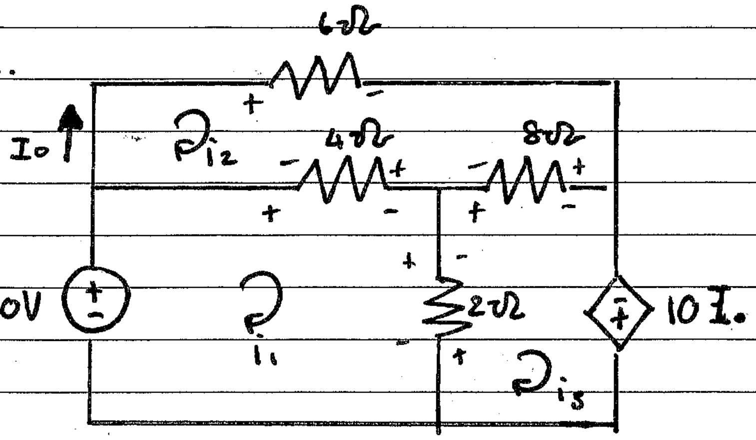
start point.

* $\sum V$ in loop $= 0$

eq.1) $-V_1 + R_1 \times I_1 + R_3(I_1 - I_2) = 0$

eq.2) $R_3(I_2 - I_1) + R_2 I_2 + V_2 = 0$

eg.2.



<u>Find I_o:</u>

$I_o = i_2$

Mesh(1): $-20 + 4i_1 + 2(i_1 - i_3) - 4i_2 = 0$

Mesh(2): $2(i_3 - i_1) + 8(i_3 - i_2) - 10 I_o = 0$

Mesh(3): $6i_2 + 8(i_2 - i_3) + 4(i_2 - i_1) = 0$

(1) $6i_1 - 4i_2 - 2i_3 = 20$
(2) $-2i_1 - 18i_2 + 10i_3 = 0$
(3) $-4i_1 + 18i_2 - 8i_2 = 0$

$$\begin{bmatrix} 6 & -4 & -2 \\ -2 & -18 & 10 \\ -4 & 18 & -8 \end{bmatrix}$$

$$i_2 = \frac{\begin{vmatrix} 6 & 20 & -2 \\ -2 & 0 & 10 \\ -4 & 0 & -8 \end{vmatrix}}{\begin{vmatrix} 6 & -4 & -2 \\ -2 & -18 & 10 \\ -4 & 18 & -8 \end{vmatrix}} \qquad \frac{1120}{-224}$$

$i_2 = -5A$
$I_o = -5A$

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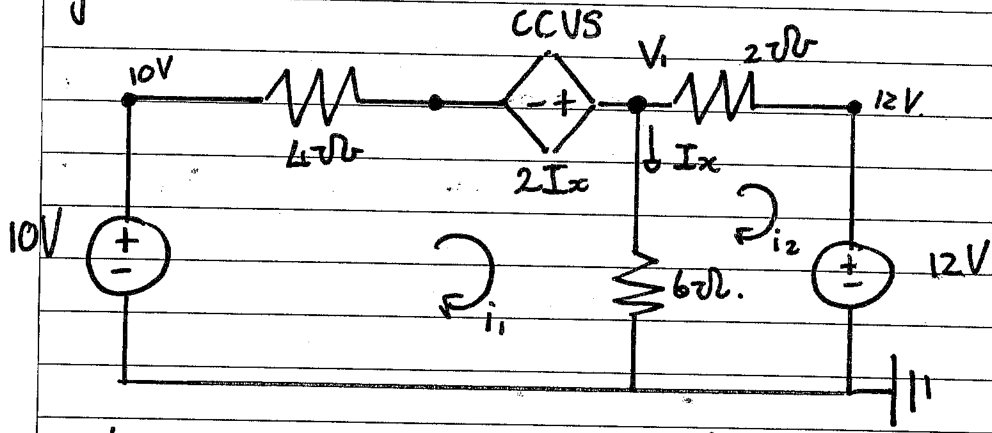
e.g.2.



~~$I_x = 6(i_2$~~

- $I_x = i_1 - i_2$

Mesh Analysis.
KVL

Mesh 1: $-10 + 4i_1 - 2I_x + 6(i_1 - i_2) = 0.$

Mesh 2: $6(i_2 - i_1) + 2i_2 + 12 = 0$

$i_1 = 0.8A \qquad i_2 = -0.9A.$

$\Rightarrow I_x = \cancel{0.1A}\ 1.7A$

Nodal Analysis:
KCL
(1 crit. node)

$$\frac{V_1 - 2I_x - 10}{4} + \frac{V_1 - 12}{2} + \frac{V_1}{6} = 0$$

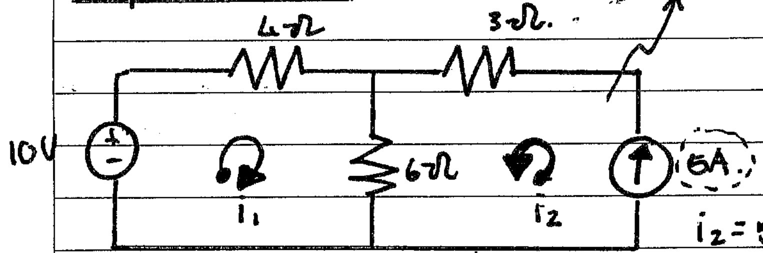
$$I_x = \frac{V_1}{6}$$

(a) sub $I_x = \frac{V_1}{6} \Rightarrow V_1 = 10.2V$

(b) sub $V_1 = 6I_x \Rightarrow I_x = 1.7A.$

Supermesh:

Both in opp. directi:

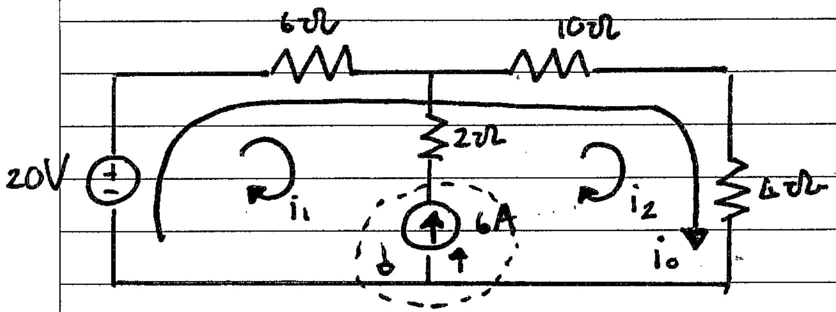


$i_2 = 5A.$

Mesh 1: $-10 + 4i_1 + 6(i_1 + 5) = 0$

$\Rightarrow i_1 = -2A.$

* ↻: $i_2 = -5A.$
* ↺: $i_2 = 5A$



① * $6 = i_2 - i_1$

NB

Mesh 1: $-20V + 6i_1 + 2(i_1 - i_2) + \ldots = 0.$

* Failed using KVL

$\Rightarrow$ Supermesh.

Supermesh: (ignore common branch)

② * $-20 + 6i_1 + 10i_2 + 4i_2 = 0$

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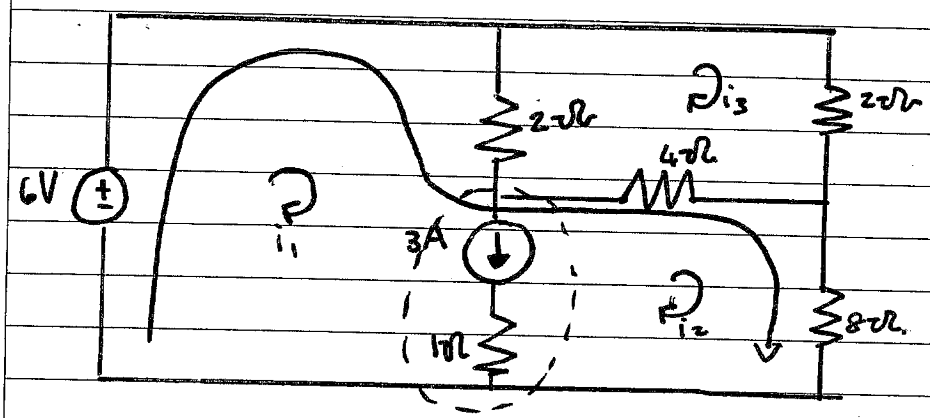
eg.3.



Branch Current: $3 = i_1 - i_2$ ①

Super mesh:

$$-6 + 2(i_1 - i_3) + 4(i_2 - i_3) + 8i_2 = 0 \quad ②$$

Mesh 3:

$$2(i_3 - i_1) + 2i_3 + 4(i_3 - i_2) = 0 \quad ③.$$

① $i_1 - i_2 - 0i_3 = 3$

② $2i_1 + 12i_2 - 6i_3 = 6$

③ $-2i_1 - 4i_2 + 8i_3 = 0$

$$\begin{bmatrix} 1 & -1 & 0 \\ 2 & 12 & -6 \\ -2 & -4 & 8 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$

* Mesh current = all mesh currents.

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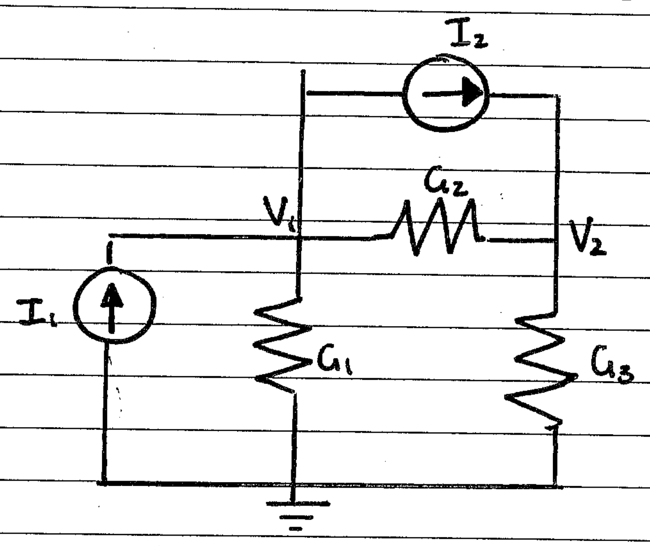
Mesh & Nodal analysis by inspection.



(2 × 2)

n × n

$C_{ji} = \underline{\underline{C_{ij}}}$, $i \neq j$

C_{11} = Total conductance (with node 1)

C_{22} = Total conductance (with node 2)

C_{12} = − (common conductance)

C_{21} = − (Common Conductance)

$$\begin{pmatrix} G_1+G_2 & -G_2 \\ -G_2 & G_2+G_3 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} I_1 - I_2 \\ I_2 . \end{pmatrix}$$

conductance matrix

nodal voltages.

vector.

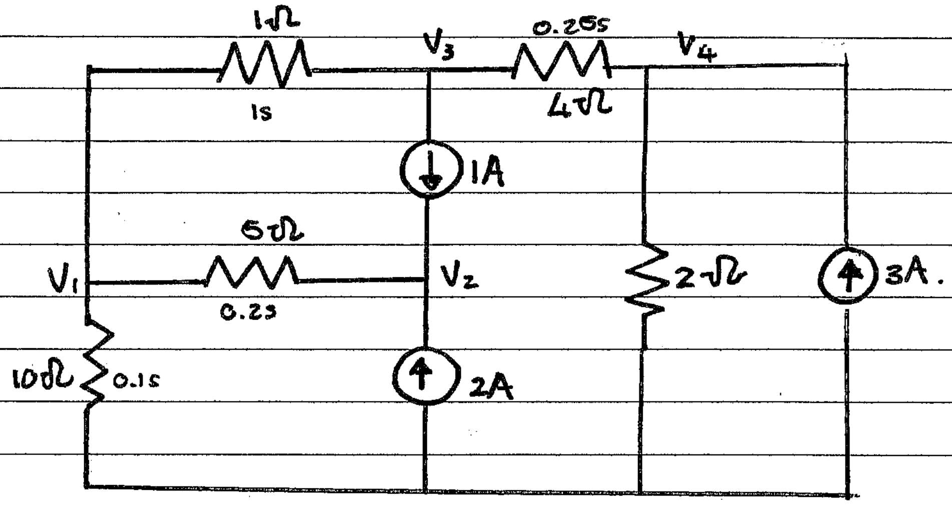
- sum of independent current source entering a node.

<u>Vector:</u>

"current account"

current entering or leaving via current source.

Nodal Analysis By Inspection:



$$\begin{pmatrix} 1.3 & -0.2 & -1 & 0 \\ -0.2 & 0.2 & 0 & 0 \\ -1 & 0 & 1.25 & -0.25 \\ 0 & 0 & -0.25 & 0.75 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \\ -1 \\ 3 \end{pmatrix}$$

*Note: it is a symetric matrix.

$C_{11} = 1+0.1+0.2 = 1.3$

$C_{22} = 0.2$

$C_{33} = 1+0.25 = 1.25$

$C_{44} = 0.75.$

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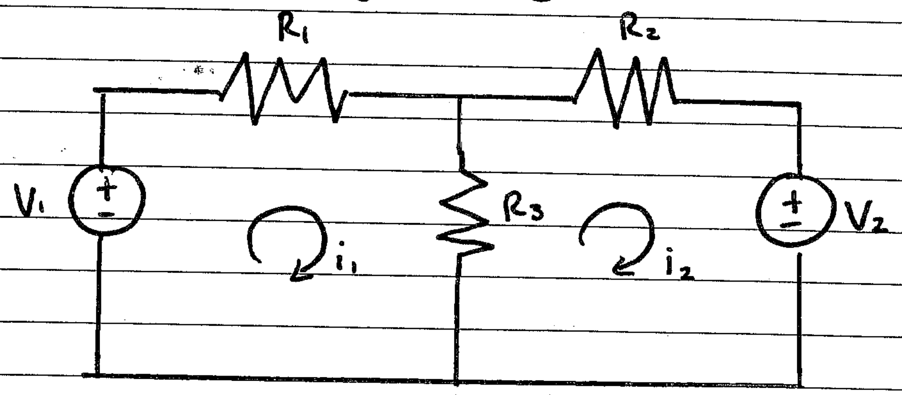
Mesh Analysis by Inspection.



$$\begin{pmatrix} R_1+R_3 & -R_3 \\ -R_3 & R_2+R_3 \end{pmatrix}\begin{pmatrix} i_1 \\ i_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ \end{pmatrix}$$

Resistance Matrix

Vector for mesh current.

sum. of independent voltage increase of a mesh.

$R_{ij} = R_{ji}$ ⟺ Symmetric.

R_{12} = Resistence between loops.$\times(-1)$

R_{21} = Resistence between loops $\times(-1)$

R_{11} = Resistence in loop

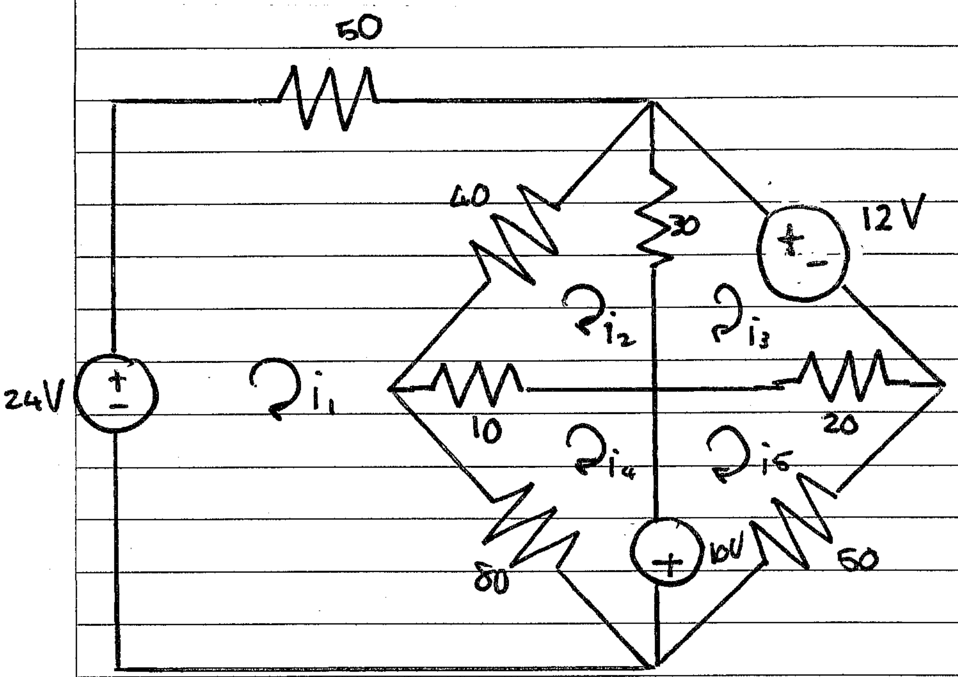
R_{22} = Resistence in loop.

note:

V_1

$\rightarrow$ (− +) $= +V_1$

$\leftarrow$ (− +) $= -V_1$



$$\begin{pmatrix} 170 & -40 & 0 & -80 & 0 \\ -40 & 80 & -30 & -10 & 0 \\ 0 & -30 & 50 & 0 & -20 \\ -80 & -10 & 0 & 90 & 0 \\ 0 & 0 & -20 & 0 & 80 \end{pmatrix} \begin{pmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{pmatrix} = \begin{pmatrix} 24 \\ 0 \\ -12 \\ 10 \\ -10 \end{pmatrix}$$

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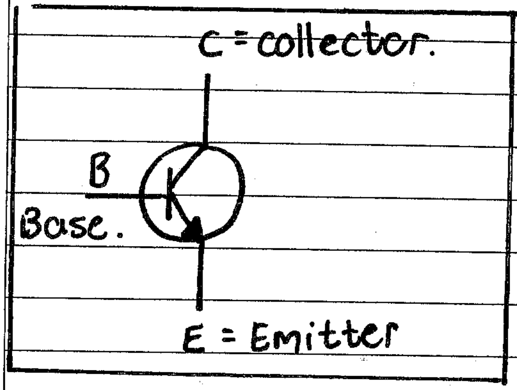
Transistors:

* BJT
  Bipolar Junction Transistor.

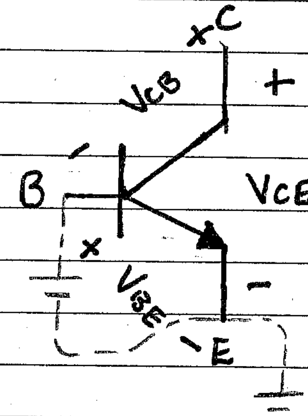


2 Types:

- BJT.
- FET.

Note: V from + to - terminal.

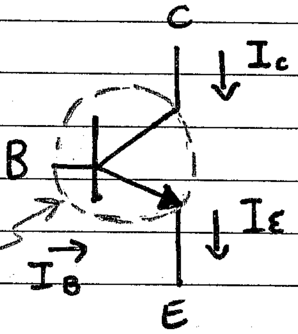
=> $\overset{-}{B}$ - - - V_{CB} - - - $\overset{+}{C}$



Voltage:

$V_{CE} = V_{BE} + V_{CB}$
$= 0.7 + V_{CB}$

$V_{BE} = 0.7V$

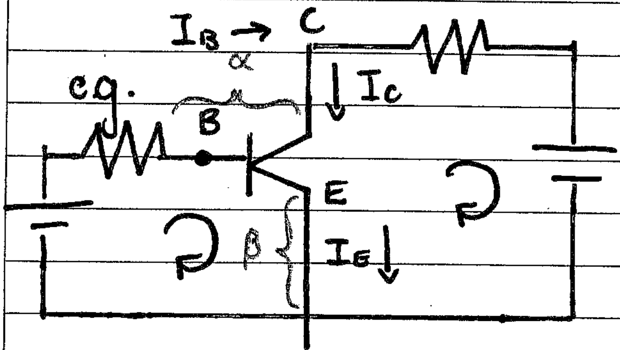


Analyse as a supernode.

Current:

$I_B + I_c = I_E$ (1)

β = common-emiter current gain.



$I_c = \beta I_B$ (2)

β :- no unit. (constant)
- 50-100

α :- aprouching 1
- 0.98-0.99.

α = common-base current gain.

$I_c = \alpha I_E$ (3)

Derived Formulas:

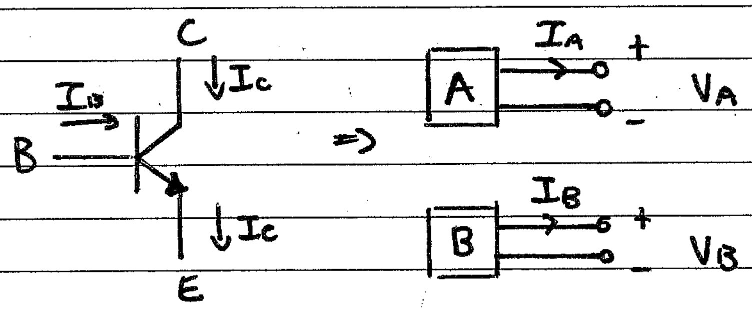
① $I_E = (1+\beta) I_B$

② $\alpha = \dfrac{\beta}{1+\beta} \Rightarrow \alpha \to 1 \; , \; \underline{\underline{\alpha < 1}}$

③ $\beta = \dfrac{\alpha}{1-\alpha}$

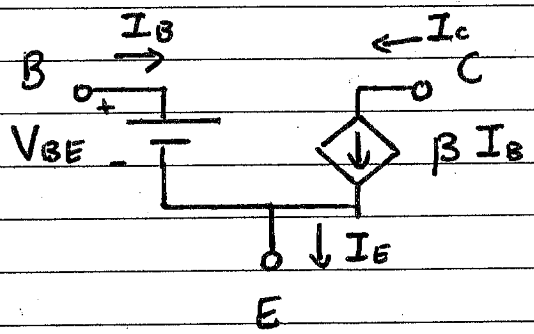
Equivalent Network model:

*Other factors were not taken into consideration.



$I_A = I_B$ & $V_A = V_B$.

=> A is equivalent to B



- $V_{BE} = 0.7V$
- $I_c = \beta I_B$
- $I_c = I_B + I_c$

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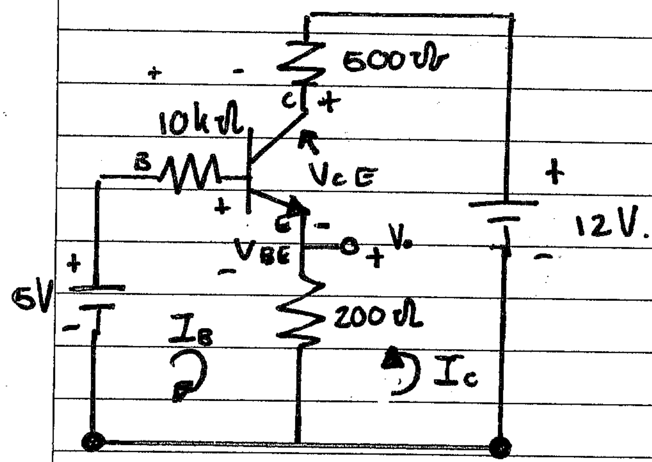
Transistor circuit analysis:



Mesh I_B: $-5 + 10000(I_B) + V_{BE} + V_o = 0$

V out : $V_o = 200(I_B + I_c)$

Mesh I_c: $-12 + 500(I_c) + V_{CE} + V_o = 0$

Relationship: $I_c = \beta I_B$

$-5 + 10^4 I_B + 0.7 + 200(I_B + \beta I_B) = 0.$

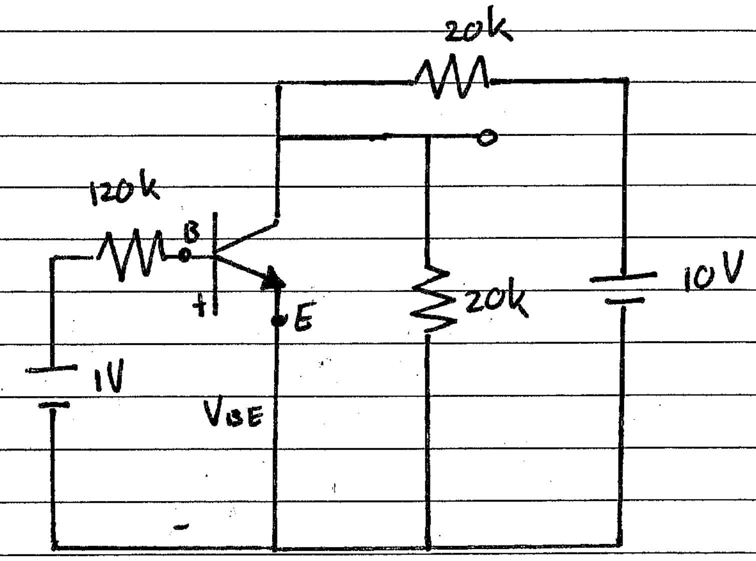
$\Rightarrow$ $10^4 I_B + 200 \times 101 I_B = 4.3$

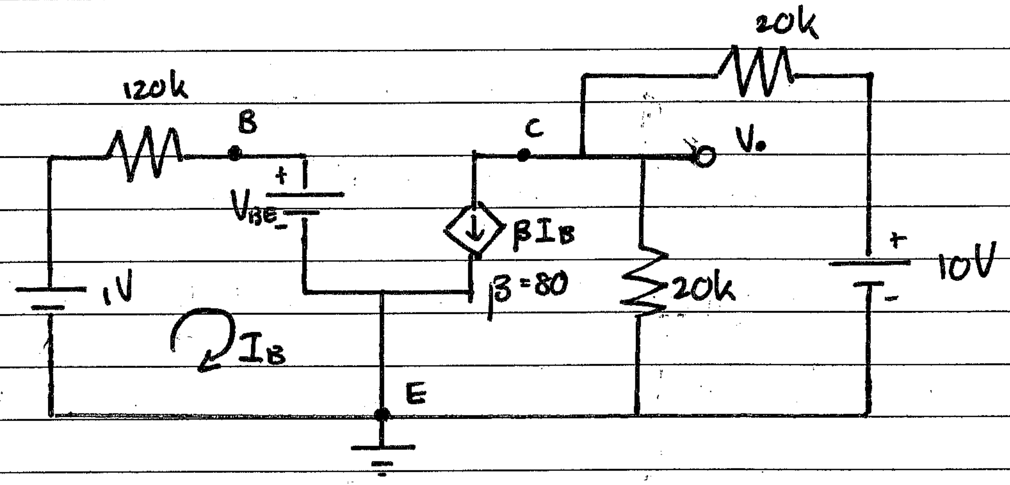
$\Rightarrow$ $30200 I_B = 4.3.$

$\Rightarrow$ $I_B = 0.1424$ mA

$-12 + 500(100 \times 0.1424 \times 10^{-3}) + V_{CE} + 200(0.14 \times 10^{-3} + 100 I_B) = 0.$

$\Rightarrow$ $V_{CE} = 2V$





$$-1 + 120\times10^{3} \times I_B + 0.7 = 0$$

$$\Rightarrow I_B = 2.5\ \mu A$$

$$\beta I_B + \frac{V_o}{20k} + \frac{V_o - 10}{20k} = 0$$

$$80 \times (2.5\times10^{-6}) \times 20\times10^{3} + 2V_o = 10$$

$$\Rightarrow 4 + 2V_o = 10$$

$$\Rightarrow V_o = 3V$$

$$I_o = \frac{3V}{20k\Omega} = 150\ \mu A$$

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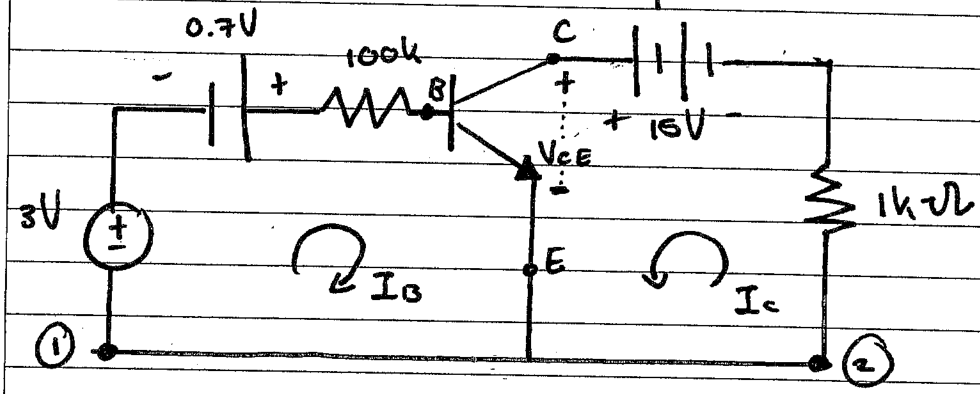
Determine: V_{CE} & I_B. $V_{BE} = 0.7$

$\beta = 100$.



Mesh 1:

$$-3 - 0.7 + (100 \times 10^3) \times I_B + V_{BE} = 0.$$

$$\Rightarrow I_B = 30\,\mu A$$

Mesh 2:

$$10^3 \times I_C - 15 + V_{CE} = 0$$

$$\Rightarrow 10^3 \underbrace{\times 100}_{\beta} \underbrace{\times 30 \times 10}_{I_B} - 15 = -V_{CE}$$

$I_C = \beta I_B$

$$\Rightarrow V_{CE} = 12V.$$

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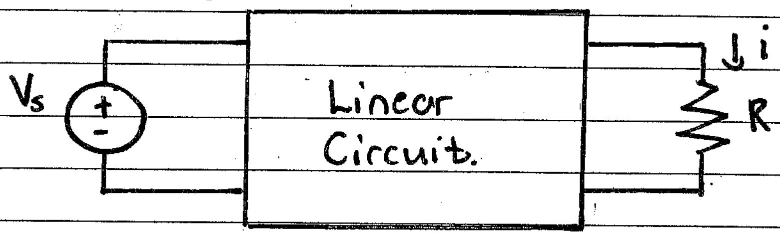
Chapter 4: Circuit Theorms.

Linearity & Superposition:



- If $V_s = 10V \Rightarrow i = 2A$ (Obseved results)

then. $V_s = 1V \Rightarrow i = 0.2$ A

$V_s = 100V \Rightarrow i = 20$ A

Two Properties: (Of linearity).

(a)* Homogeneity (scaling)

* $V = I \times R$ $\quad$ $kV = kI \times R$

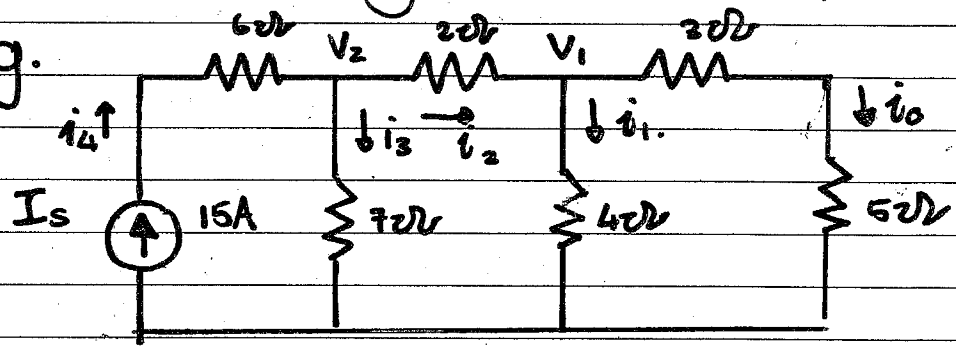
(b)* Additivity

* $\left.\begin{array}{l} V_1 = I_1 R \\ V_2 = I_2 R \end{array}\right\} \Rightarrow V_1 + V_2 = (I_1 + I_2) R$

Linearity:

Output linearly related to Input.

e.g.



$V_1 = 1 \times (3+5) = 8V$ $\quad$ $V_2 = V_1 + I_2 \times 2 = 14\ V$

$I_1 = 8/4 = 2A.$ $\quad$ $I_3 = 14/7 = 2A.$

$I_2 = I_1 + I_o = 3A.$

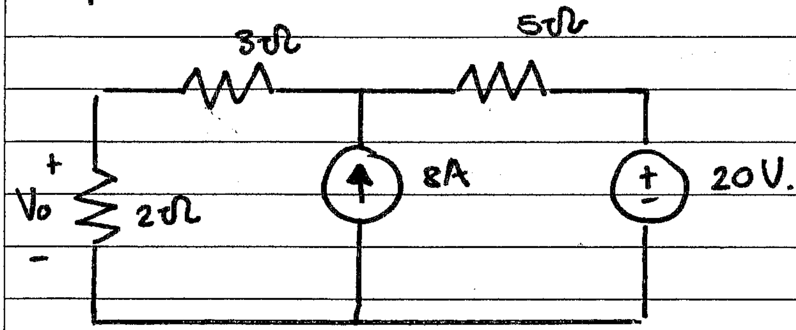
Note:

Let $i_o = 1A$.

Find $\dfrac{i_o}{i_s} = k.$

$I_{o_{actual}} = k \cdot I_{s_{actual}}.$

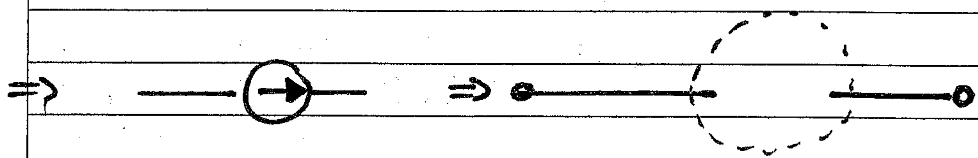
Superposition Theorem:



1. Turn off INDEPENDENT sources.

Voltage Source:
- short circuit.

Current Source:
- Open circuit.



* Leave Dependent Sources.

2. Repeat step 1 with analysis.

3. Summation (Additivity).

Theory: Voltage across element is the sum of voltage across element due to EACH independent source alone.

Short Circuit: $R = 0\,\Omega$
Open Circuit: $R = \infty\,\Omega$.

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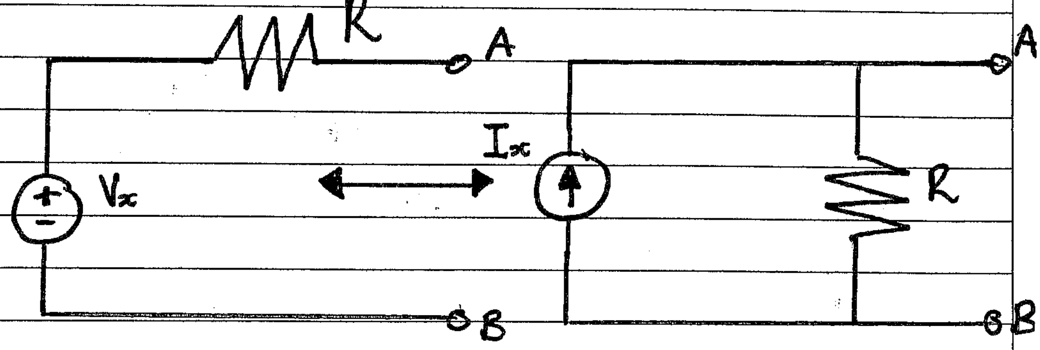
Source Transformation:

Equivalent circuit: A new circuit whose V & I characteristics are identical with the original circuit.



* Measurements at A & B yields same result.

Value:

① Change Configuration : (series → parallel) (V ↑ series, A ↑ parallel)

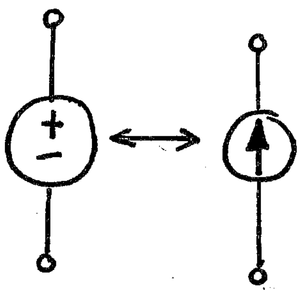
② Define Values : ⟶ R (does NOT change)

③ Value of Source: $I_s = \dfrac{V_s}{R}$ OR $V_s = I_s R$

Direction:

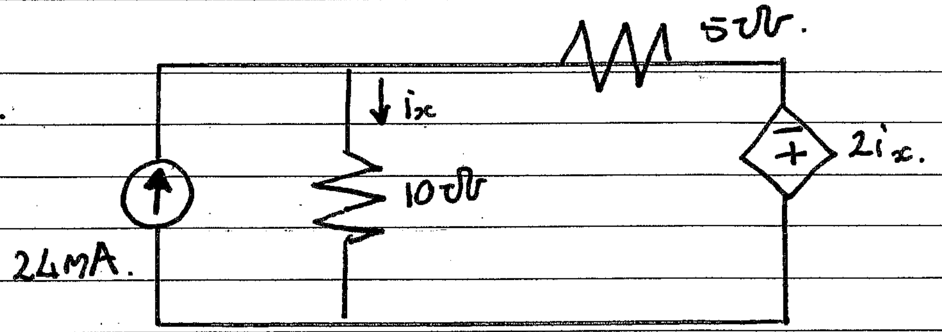
① Arrow of Ⓐ pointing to + end.

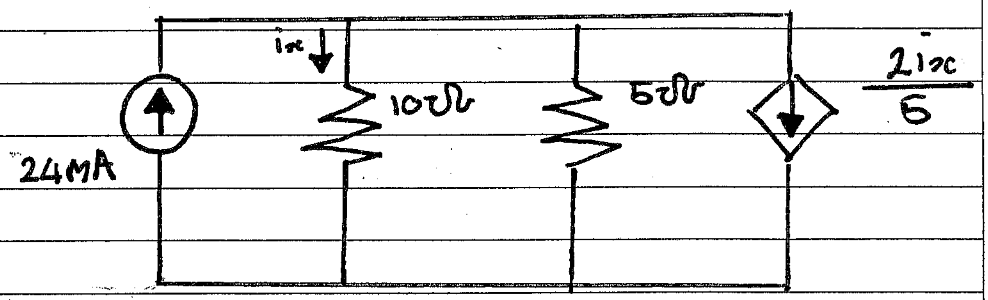
i.e.



* Same applies to independent & dependent sources.

e.g.1.





$$i_x = \frac{5}{10+5}\left(\frac{24MA - 2i_x}{5}\right)$$

Notes:

Do NOT transform part of circuit with unknown variable.

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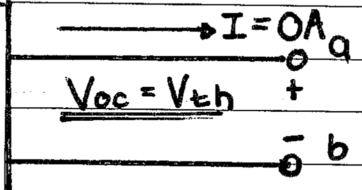
* Transistor NOT linear.
* Linear : R, Voltage s. Current s.

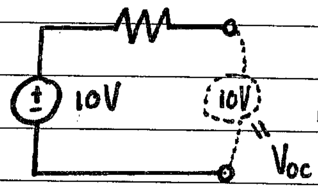
Thévenin's Theorem:

Linear two-di-terminal circuit.



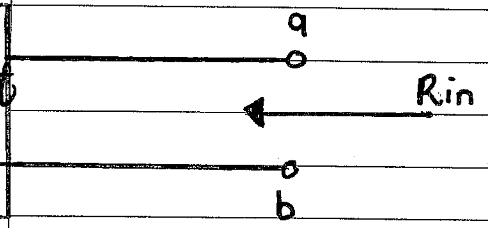
Voc = Open Circuit Voltage

e.g.



$V_{th} = 10V$
$R_{th} = 5\,\Omega$

Linear circuit with all independent sources set equal to zero

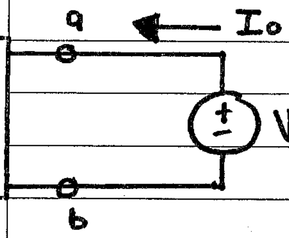


$R_{th} = R_{in}$

Method 1: Zeroing of sources :
* If the circuit has no dependent sources.

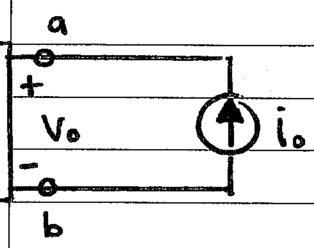
Method 2: Application of test sources :
* If the circuit DOES contain dependent sourc

Circuit with all independent sources set = 0



$$\boxed{R_{Th} = \frac{V_o}{I_o}}$$

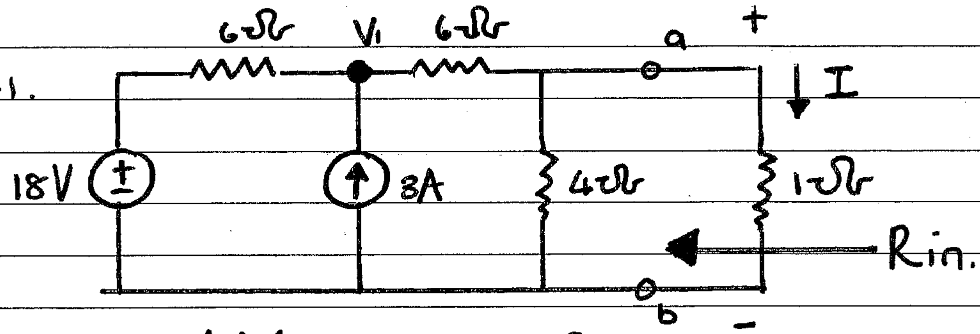
Circuit with all independent sources set = 0



$$R_{Th} = \frac{V_o}{i_o}$$

* Note: We ignore the 1 Ω resistor.

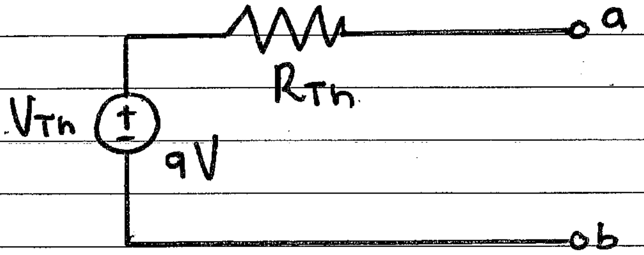
e.g. 1.



$V_{Th} = V_{ab}$

↳ (Use diff. methods)
(V over 4 Ω)

- Nodal
- Mesh
- Superposition.
- Transformation.



<u>For V_{Th}:</u>

$$\frac{V_1 - 18}{6} + \frac{V_1}{10} = 3$$

$$90 = 5V_1 - 90 + 3V_1$$

$$V_1 = 22.5\text{ V}$$

$$V_{Th} = V_1 \frac{4}{10} \qquad \underline{\underline{V_{Th} = 9\text{ V}}}$$

<u>For R_{Th}:</u>

$$R_{Th} = 4 \parallel (6+6) = \frac{48}{16} = \underline{\underline{3\,\Omega}}$$

<u>Notes:</u>

<u>Calculating V_{TH}.</u>

* Remove load.
* Calc. voltage drop across terminal.

<u>Calculating R_{TH}:</u>

* Short all voltage sources.
* Open all current sources.
* Circuit with ONLY resistors.
* Calc. $R_{eq} = R_{TH} = R_{in}$.

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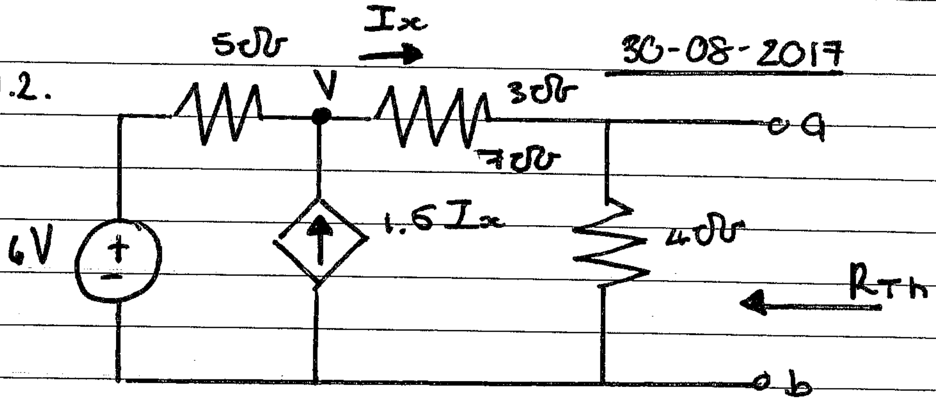
30-08-2017

e.g.2.



$V_{Th} = V_{ab}$, $\frac{V-6}{5} + \frac{V}{7} = 1.5 I_x = 1.5 \frac{V}{7}$

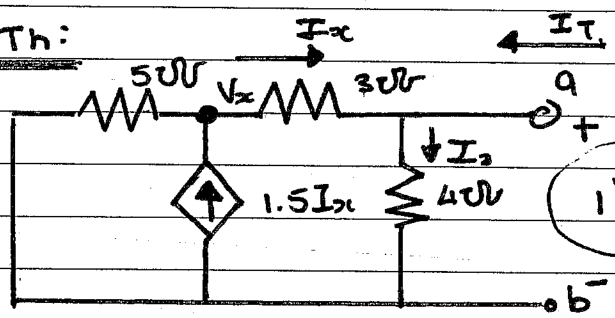
$\Rightarrow \frac{V-6}{5} = \frac{V}{14}.$

$\Rightarrow V = 28/3\ V$

For V_{Th}:

$V_{Th} = \frac{28}{3} \times \frac{4}{7} = \frac{16}{3}\ V$

For R_{Th}:



introduce a test voltage.

(1) $1.5 I_x = \frac{V_x}{5} + \frac{V_x - 1}{3}$

(2) $\frac{V_x - 1}{3} = \frac{V_x}{5}$

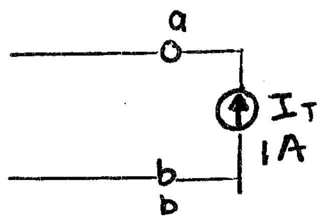
$V_x = -5V$

$I_x = -2A$

(3) $I_2 = \frac{1}{4} = 0.25A$

$I_T = 2 + 0.25 = 2.25A.$

$R_{Th} = \frac{V_{Th}}{I_T} =$



same circuit diff. approach:

$$0.5 I_x = \frac{V}{6} = 0.5 \cdot \frac{V - V_T}{3}$$

$$\frac{V}{6} = \frac{V - V_T}{6} \quad (1)$$

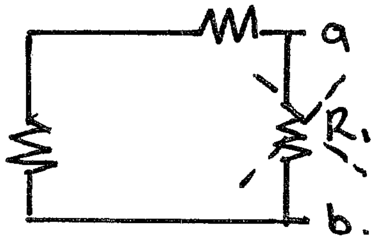
$$\frac{V - V_T}{3} + 1 = \frac{V}{4} \quad (2)$$

$$V_T = \frac{4}{9} V.$$

$$R_{Th} = \frac{V_{Th}}{I_{Th}} = \frac{1}{9/4} = \frac{4}{9}\ \Omega$$

Notes:

Cancel out the R_1 resisto if over a-b terminal.



* If Dependent sources present use a test voltage.

$$R_{TH} = \frac{V_o}{i_o}.$$

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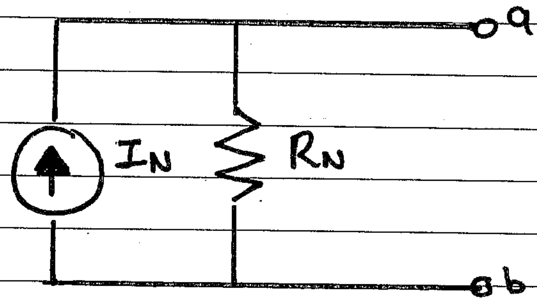
30-08-2017.

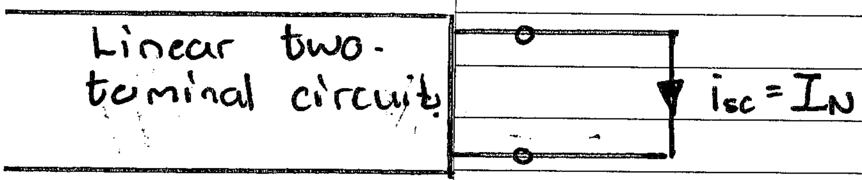
(chap4)

- Norton's eq. circuit = Thevenin eq. circuit.
- $R_{Th} = R_N$
- $I = \frac{V_{Th}}{R_{Th}}$
- $I = I_N$.

Norton's Theorem: (4.6).



Determine I_N:



- Norton equivalent current is the short circuit current.

Determine R_N:

Method 1: Zero independent sources.

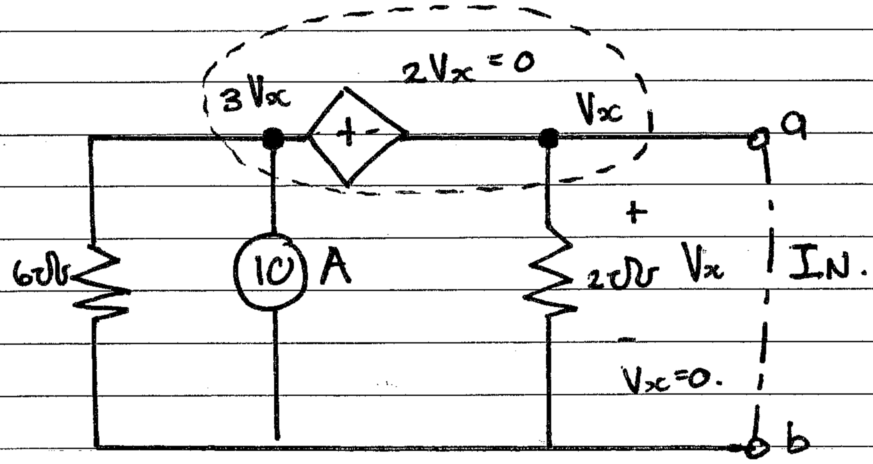
Method 2: by testing → (A), (V)

Method 3:

$R_{Th} = \frac{V_{oc}}{I_{sc}}$

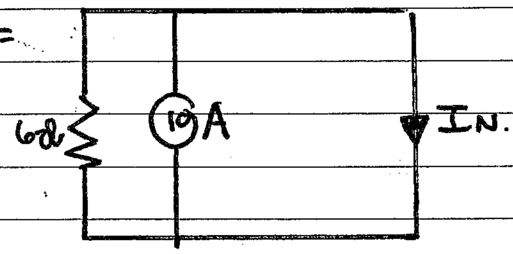
$R_{Th} = R_N =$

eg.



I_N:

$I_N =$



$\underline{I_N = 10A}$.

R_N:

$$R_N = \frac{V_{oc}}{I_N}, \quad I_N = 10A$$

$$10 = \frac{3V_x}{6} + \frac{V_x}{2} \Rightarrow V_x = 10V$$

$$R_N = 1\Omega$$

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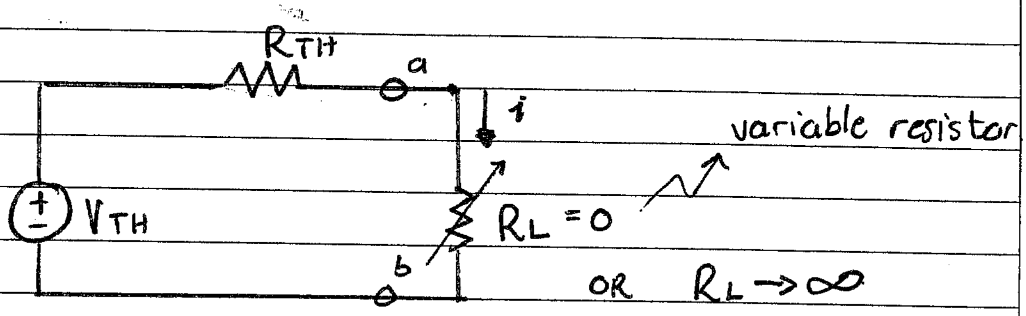
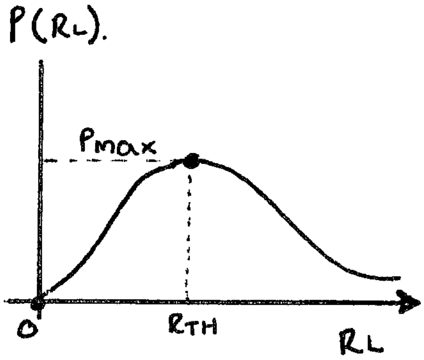
6-09-2017

Maximum Power Transfer.



Power transfer profile with diff. R_L:



$$P(R_L) = i^2 \cdot R_L = \left(\frac{V_{TH}}{R_{TH}+R_L}\right)^2 \cdot R_L$$

if $R_L \to \infty$, $i = 0$ (open circuit)

$$P_{max} = \left(\frac{V_{TH}}{2R_{TH}}\right)^2 \cdot R_{TH} \quad \{\text{Where } R_L = R_{TH}\}$$

$$* \quad P_{max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{V_{RL}^2}{R_{TH}}$$

Proof:

Note: $f(x) = \begin{cases} \max \\ \min \end{cases}$

$f'(x) = 0$.

$$P'(R_L) = \left(V_{TH}^2 (R_{TH} + R_L)^{-2} R_L\right)'$$

$$= V_{TH}^2 \left[-2(R_{TH}+R_L)^{-3} R_L + (R_{TH}+R_L)^{-2}\right]$$

$P_{max} \to P'(R_L) = 0$

$$\Rightarrow \frac{-2R_L}{(R_{TH}+R_L)^3} + \frac{1}{(R_{TH}+R_L)^2} = 0.$$

$R_{TH} + R_L \neq 0$

$\Rightarrow -2R_L + R_{TH} + R_L = 0.$

$\Rightarrow R_L = R_{TH} \longrightarrow P'(R_L) = 0.$

$\longrightarrow$ Maximum power.

EBN 122. # Operational Amplifiers: 9-09-2017.

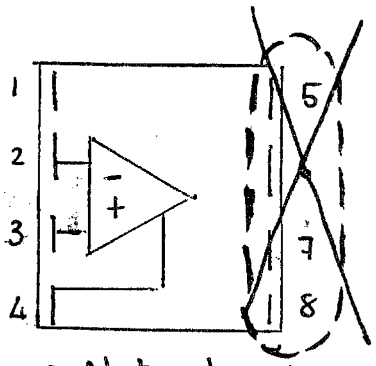
Chapter 5:

Op Amp:

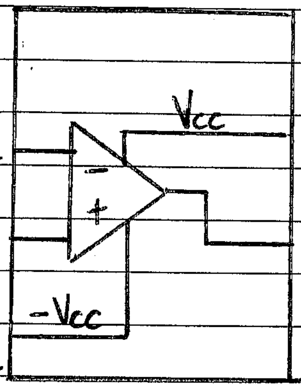
* Behaves as a VCVS.

*Oporational Amplifyer.

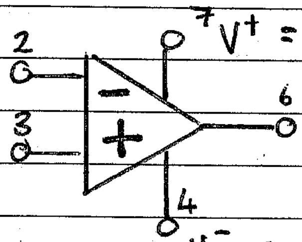
- integrated circuit. (IC).
- active ~~current~~ element. circuit.



- Note how incorrect nrs. causes error.



- Balance 1 | 8 • No connection.
- Inverting input. 2 | 7 • V^+
- Noninverting input 3 | 6 • Output.
- V^- 4. | 5 • Balance.



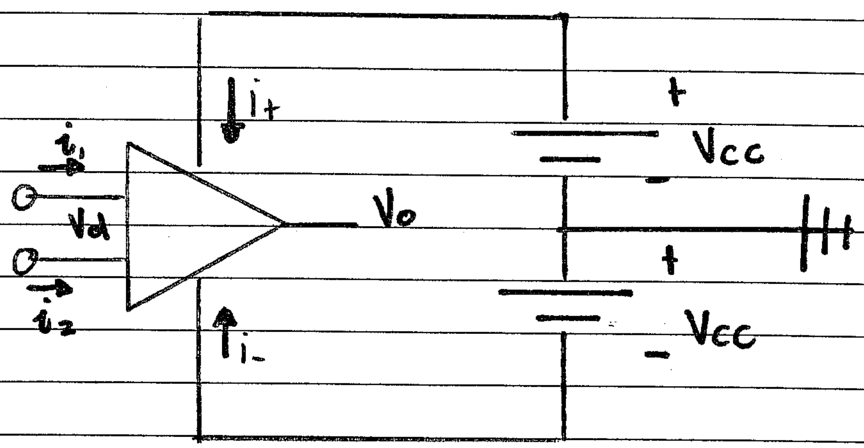
- Inverting Input.
- Noninverting Input.

$^7 V^+ = +V_{cc}$.

^{6} Output.

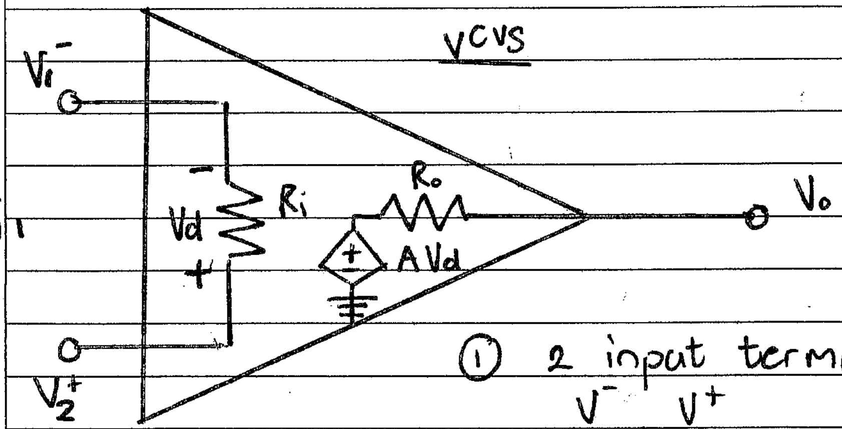
$^4 V^- = -V_{cc}$

Connection:



Equivalent Circuit: 1

$V_d = V_2 - V_1$



1. 2 input terminals. V^-, V^+
2. R_i : input resistance.
3. R_o : output resistance.

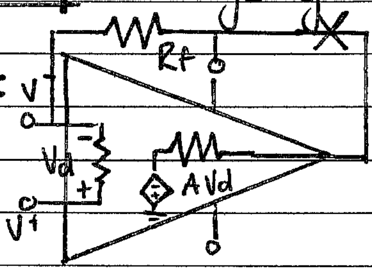
* $V_d = V^+ - V^-$ (voltage diff. of input)

OR.
$V_o = A(V_2 - V_1)$

* $V_o = AV_d$ (function of V_d)

* A : **Open loop** voltage gain.

* Open loop :



feedback loop.

• Op Amps amplifies the voltage difference of V^+ & V^-

Closed Loop Gain $= \dfrac{V_o}{V_s}$.

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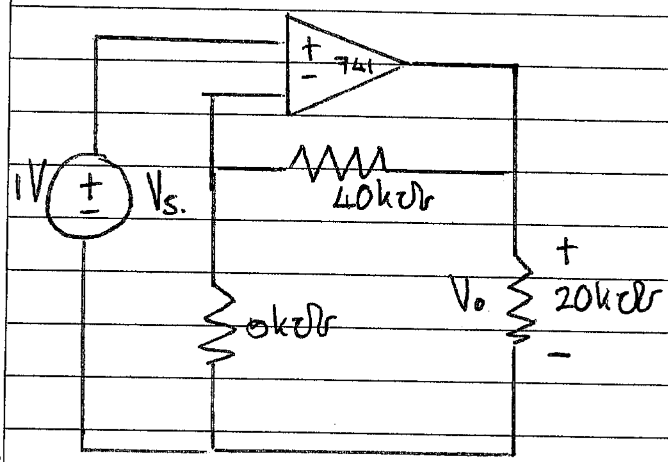
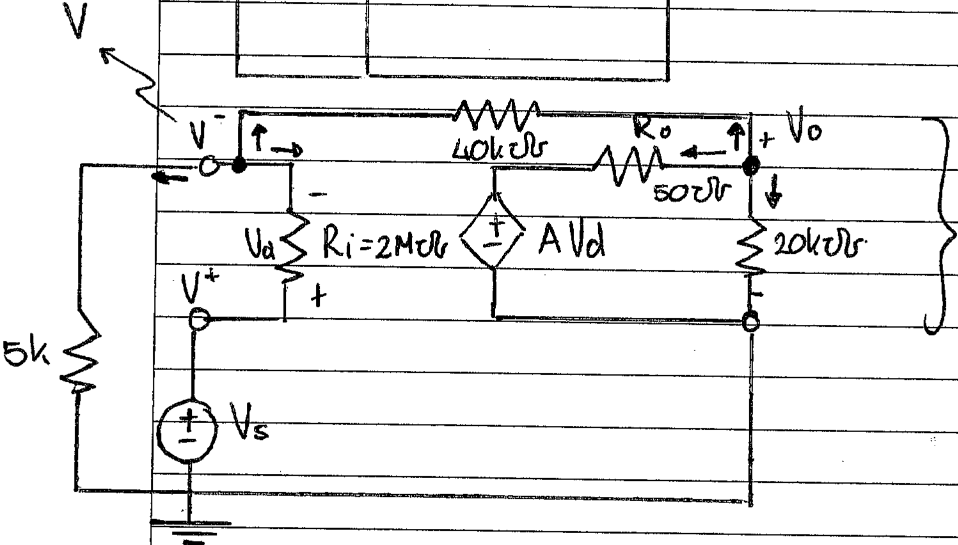
6-09-2017

Example: $A = 2\times10^5$ 741 Op Amp.

$R_i = 2M\Omega$

$R_o = 50\Omega$





Equivalent Circuit.

① $\dfrac{V}{5k} + \dfrac{V-V_o}{40k} + \dfrac{V-V_s}{2000k} = 0$

② $\dfrac{V-V_o}{40k} = \dfrac{V_o - AV_d}{R_o} + \dfrac{V_o}{20k} = 0$

1) $400V + 50V - 50V_o + V - V_s = 0.$ $\quad V = \dfrac{50V_o + V_s}{451}$

$451V - 50V_o = V_s \quad \Rightarrow$

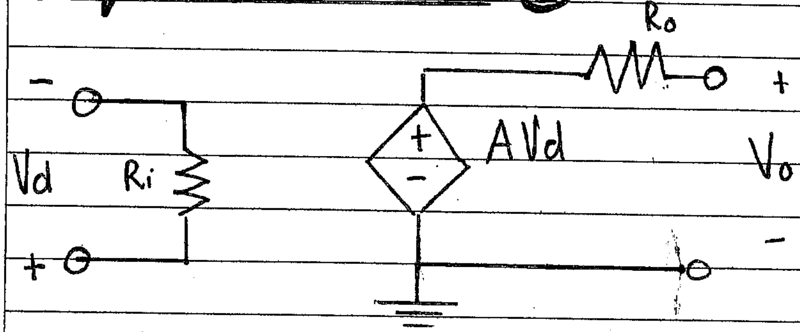
2) $V - V_o = (40\times10^3) \cdot \dfrac{V_o - 2\times10^5(V_s - V)}{50} + 2V_o$

$V - V_o = 800V_o - 16\times10^7(V_s - V) + 2V_o$

$\Rightarrow V = \dfrac{50V_o + V_s}{451}$, $V_s = 1V$.

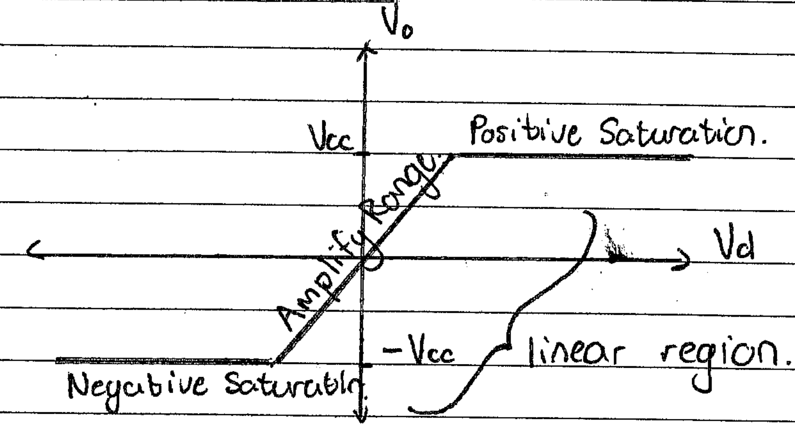
V_{cc} = Supply Voltage.

Equivalent Circuit ②.



Parameter	Range	Ideal
A	10^5 – 10^8	∞
R_i	$10^6 - 10^{13}\,\Omega$	∞
R_o	$10 - 100\,\Omega$	0
V_{cc}	5 - 24 V	

Practical limitations:



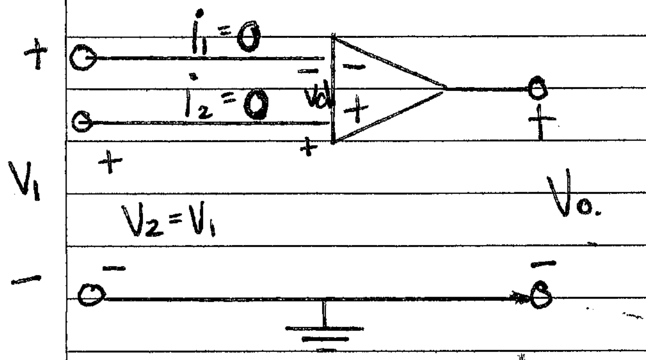
Linear region: $V_d = V^+ - V^-$

$V_o = AV_d$

Positive saturation: $V_o = V_{cc}$ (max output).

Negative saturation: $V_o = -V_{cc}$

Ideal Op Amp:



since $R_i = \infty$
(a) $i_1 = i_2 = 0$.
(b) $V_d = V_2 - V_1$
(c) $V_2 = V_1$

Characteristics:

1. Infinite open-loop gain, $A \approx \infty$
2. Infinite input resistance, $R_i \approx \infty$
3. Zero output resistance, $R_o \approx 0$.

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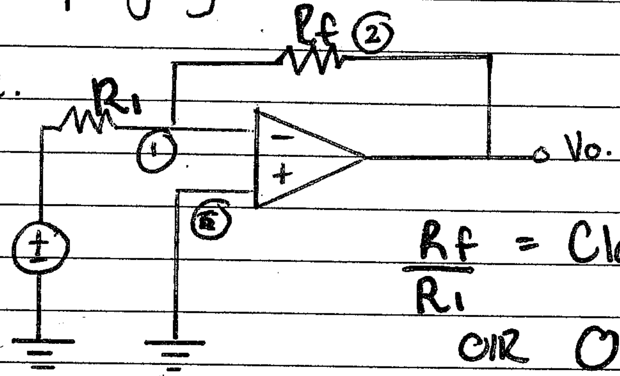
Inverting & Noninverting Amps.

Inverting Amp.

1. Config. of inverting Amp
   a) Signal input from inverting terminal.
   b) Only 1 source connected to V^-
   c) Non-inverting input is grounded.
2. Function:
   - Reversing polarity of input signal while amplifying it.

i.e.



$\frac{R_f}{R_1}$ = Closed-loop gain. OR Op-amp gain.

Non-inverting Amplifier:

* Produce +ve voltage gain.
* Source connected to V^+

Summing amplifier:

* More than 1 source.

Difference amplifier:

* Source at each input.

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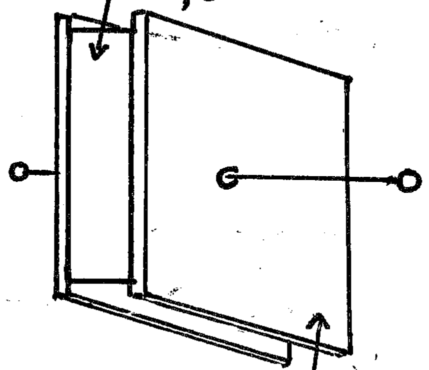
Chapter 6: Capacitors:



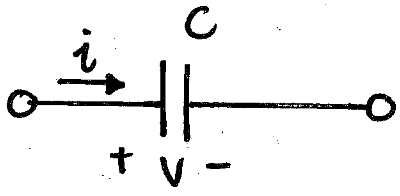
- Passive Element.
- Store energy in electric field.

$$C = \frac{\varepsilon A}{d}$$

A = Surface Area, plate $[m^2]$
ε = Permittivity, dielectric [F/m]
d = Distance between plates [m]

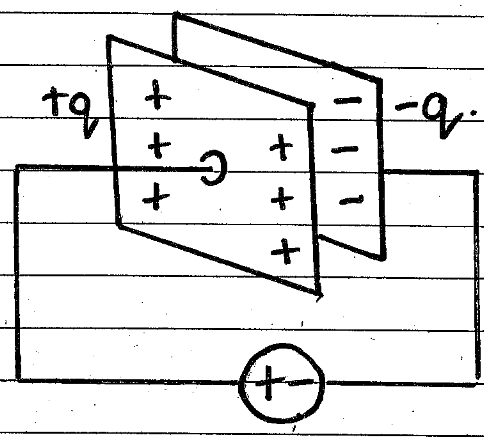
$\varepsilon = \varepsilon_r \cdot \varepsilon_o$

ε_r = Relative permittivity of material.
ε_o = Permittivity of air (8.854×10^{-12} F/m)



C = Capacitance
↳ Farad, [F]

Electrical Characteristics:



$$C = \frac{q}{V}$$

$q(t) = C \times V(t)$

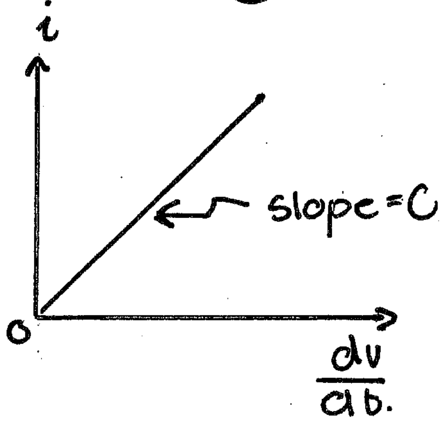
$$\boxed{i(t) = C \frac{d(V(t))}{dt}}$$

$i > 0$ charging.
$i < 0$ discharging.

$$\boxed{V(t) = \frac{1}{C}\int_{-\infty}^{t} i(t)\,dt = \frac{1}{C}\int_{t_o}^{t} i(t)\,dt + V(t_o)}$$

Rhyme: qualification = CV.

$$i = C \times \frac{dv}{dt}$$



Energy: (w), [J]

① $$w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$$

* energy stored in capacitor.

Power:

② $$p = vi = v \times C \times \frac{dv}{dt}$$

Characteristics:

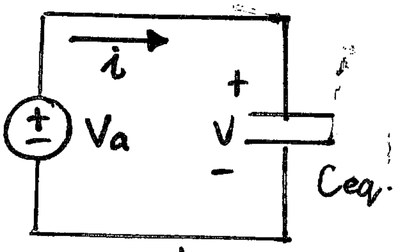
$\left(\frac{dv}{dt} = 0\right)$[DC]

* Open Circuit for DC

if $\frac{dv}{dt} = \infty$ $i = \infty$
=> ∞ heat
i.e. can't change abruptly.

* Voltage cannot change abruptly.
* Does NOT dissapate energy, only stores energy (charging) or provide energy (discharging)

[redacted] · EBN 122. 20-09-2017.



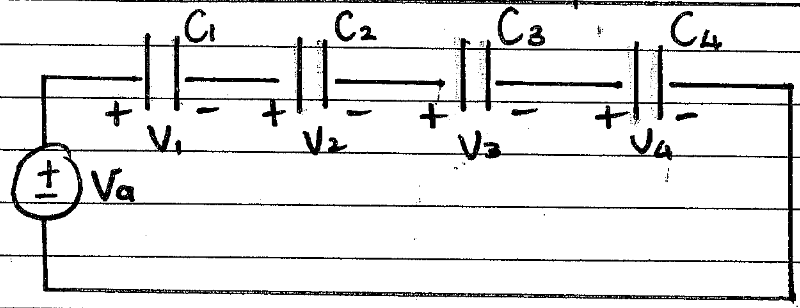
$$V = \frac{1}{C}\int_{t_0}^{t} i\,dt + V(t_0)$$

$$i = C\frac{dv}{dt}$$

* Series Capacitors add like ∥ Resistors.

* Parallel Capacitors add like series Resistors.

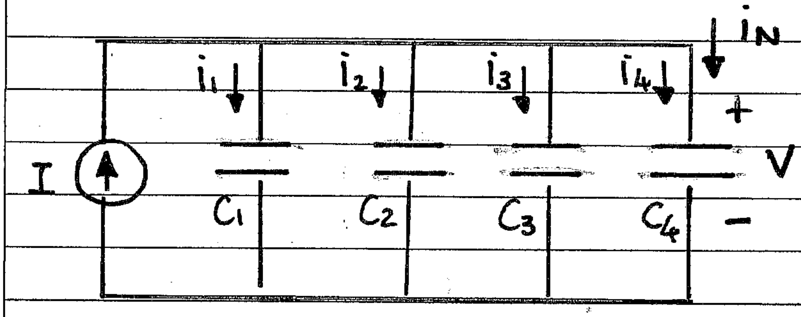
Series & Parallel Capacitors:

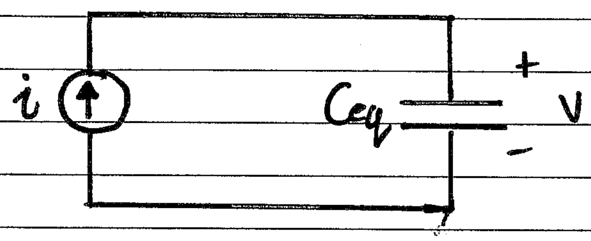


- All capacitors in series with V-source.
- $V_1 + V_2 + V_3 + V_4 = V_a$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \frac{1}{C_4}.$$

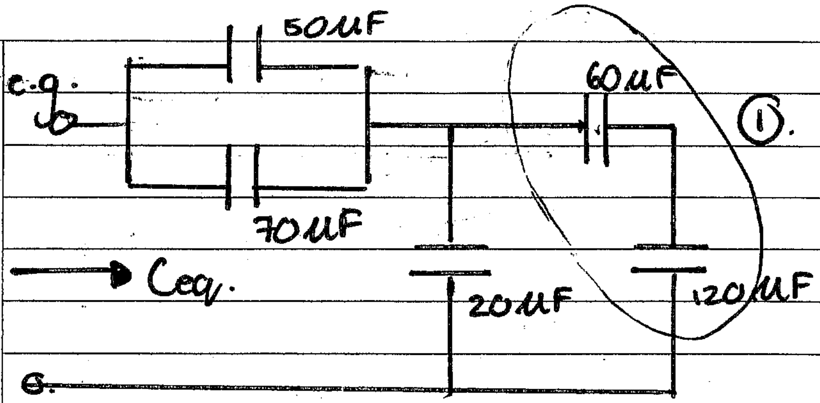
$$V = \left(\frac{1}{C_{eq}}\right)\int_{t_0}^{t} i\,dt + V(t_0).$$





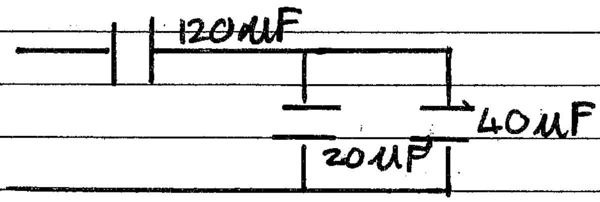
$$C_{eq} = C_1 + C_2 + C_3 + C_4$$

$$i = C_{eq}\frac{dv}{dt}.$$



① $\dfrac{60 \times 120}{60+120} = 40\mu F$

② $120\ \mu F$



$$C_{eq} = \frac{60 \times 120}{60+120}\ \mu F = 40\mu F.$$

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Examples:

1) $C = 3\mu F$, calc. Voltage.
$q = 0.12 mC$ How much w stored?

$W = \frac{1}{2} CV^2$ $\quad W = \frac{1}{2} C \left(\frac{q}{c}\right)^2$

$q = CV$

$W = \frac{1}{2} \frac{(0.12\times10^{-3})}{3\times10^{-6}}$

$W = 2.4\,mJ$ $\quad V = \frac{q}{C} = \frac{0.12\times10^{-3}}{3\times10^{-6}}$

2) $C = 10\mu F$
$V(t) = 50\sin(2000t)$ V
calc. $i(t)$.

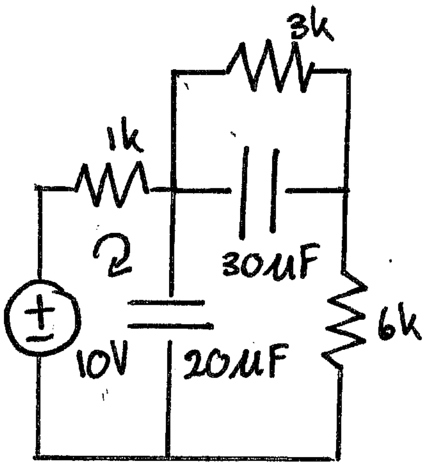
$i(t) = C \times \frac{dv(t)}{dt}.$
$= (10\times10^{-6}) \times 50 \times 2000 \times \cos(2000t).$

3) $i(t) = 50\sin(120\pi t)$ mA
$C = 100\mu F$
calc. $V(1ms)$ & $V(5ms)$ Assume $V(0) = 0$

$$V(t) = \frac{1}{C}\int_0^t 50\sin(120\pi t)\times10^{-3}\, dt$$

$$V(t) = \frac{50}{12\pi}\left(-\cos(120\pi t) + 1\right)$$

$V(1ms) = 93.137\,mV$



DC conditions.

Calc. W in 2 capacitors.

* fully charged.

4) $W = \frac{1}{2} CV^2$ $V_{30\mu F} = 3V$

$V_{20\mu F} = 9V.$

$W_{30\mu F} = \frac{1}{2}(30 \times 10^{-6}) \times 3^2$

$= 135\,\mu J$

$V_{30\mu F} = \frac{3k}{1k + 3k + 6k} \times 10.$

KVL: $-10V + \frac{1k}{10k} \times 10 + V_{20\mu F}$

$V_{20\mu F} = 9V.$

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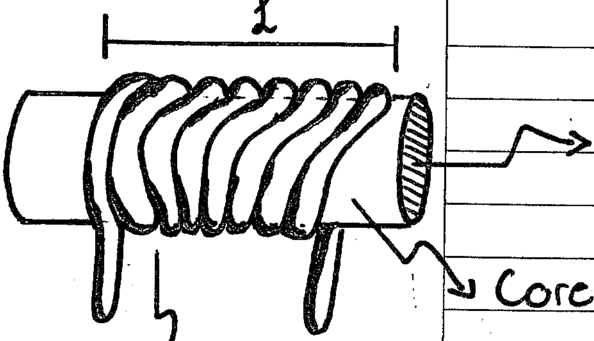
(Chapter 6).

Inductors:

- Coil of conducting wire.

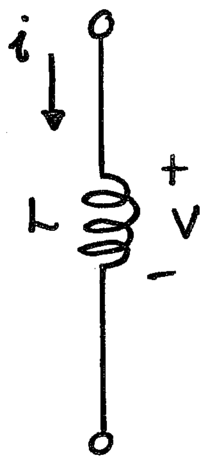
* Passive element
* Store energy in magnetic field.



$$L = \frac{N^2 \mu A}{l} \quad [H]$$

Inductance, L [H]. (Henry)

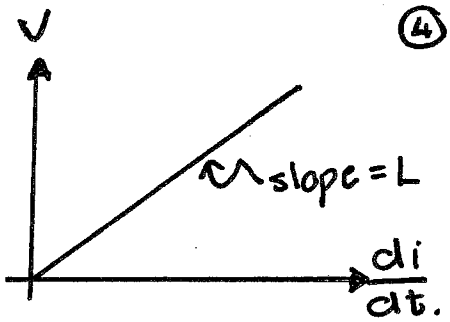
N : Number of turns.
μ : Permeability of core [H/m].
A : Cross-sectional area [m^2]
l : Length of coil [m]
μ_r : Relative permeability (scalar).
μ_0 : Permeability of vacuum [H/m]



① $i = \frac{1}{L} \int_{t_0}^{t} v(t)\, dt + i(t_0)$

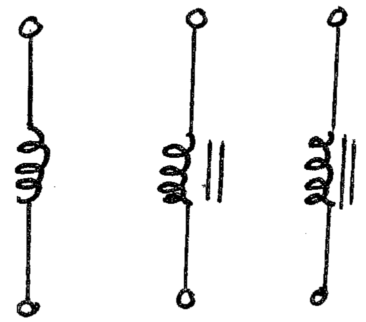
② $v = L \frac{di}{dt}$ ③ $p = vi = L \frac{di}{dt} i$ (Power).

④ $w = \frac{1}{2} L i^2$ (Energy)



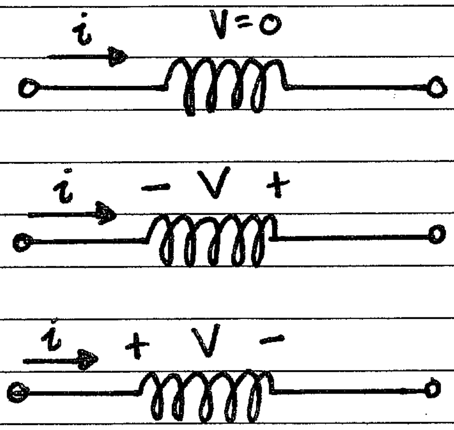
* Inductor opposes change in current.

Metal core



Hollow (air) core.

Variable



i = constant

i = increasing.

i = decreasing.

Electrical Characteristics. (Inductors)

$\left(\frac{di}{dt} \neq \infty\right)$

* Short circuit to DC $\left(\frac{di}{dt} \neq 0\right)$

* Current can't change abruptly.

* Does NOT dissapate energy.

Short Circuit in DC

e.g. Calc. V through 1mH inductor if $i(t) = 20\cos(100t)$ mA.

$$E = \frac{1}{2} L\, i(t)^2 = \frac{1}{2} \times 10^{-3} \times \left[20\cos(100t) \times 10^{-3}\right]^2$$

$$= 200\cos^2(100t) \times 10^{-9}$$

$$= 0.2\cos^2(100t)\ \mu J.$$

$$V = L\frac{di}{dt} = 10^{-3}(20\cos(100t) \times 10^{-3})$$

$$= -2\sin(100t)\ mV.$$

e.g. V across 2H inductor is $v(t) = 10(1-t)$ V. Determ. i through inductor at $t=4$ if $i(0) = 2A$.

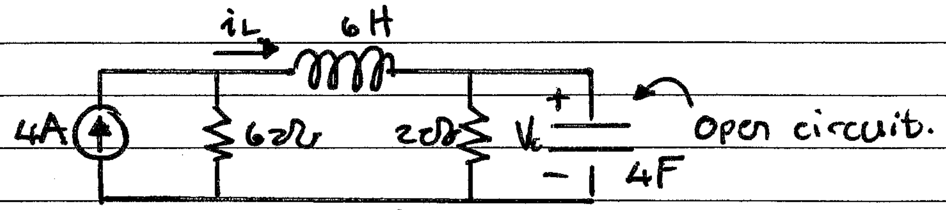
$$i(t) = \frac{1}{L}\int_0^t 10(1-t)\,dt + i(0).$$

$$= 5\int_0^t (1-t)\,dt + i(0).$$

$$= 5\left[t - \tfrac{1}{2}t^2\right]_0^t + 2.$$

$$i(4) = 5\left[4 - \tfrac{1}{2}(16)\right] + 2 = -18A.$$

e.g.



$$i_L = 4 \times \frac{6}{6+2} = 3A$$

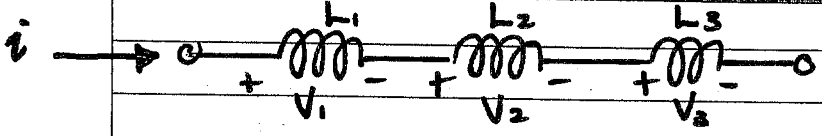
$$V_c = 3 \times 2 = 6V$$

$$E_c = \frac{1}{2} C V_c^2 = \frac{1}{2} \times 4 \times 6^2 = 72J.$$

$$E_L = \frac{1}{2} L\, i$$

Series & Parallel inductors:

Series.

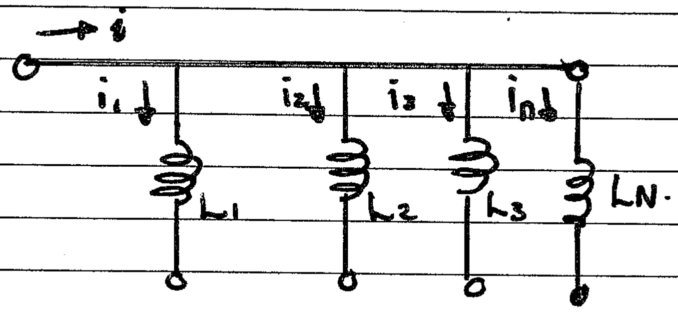


$$V = V_1 + V_2 + \ldots + V_N$$

* Add same as resistance.

$$L_{eq} = L_1 + L_2 + \ldots + L_N.$$

Parallel.



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \ldots + \frac{1}{L_N}$$

$$i = \frac{1}{L}\int_{\infty}^{t} v(t)\,dt + i(t_0)$$

Resistance.

(Ω) Ohms.

Inductance

(H) Henries.

Capacitance

(F) Farads.

Current, Voltage Relationship:

Element	Inductance	Capacitance.
V	$v = L \frac{di}{dt}$	$v(t) = \frac{1}{C} \int_{t_0}^{t} i(t)dt + V(t_0)$
i	$i = \frac{1}{L} \int_{t_0}^{t} v(t)dt + i(t_0)$	$i(t) = C \frac{dv(t)}{dt}$
P	$p = Li\left(\frac{di}{dt}\right)$	$p = Cv\left(\frac{dv}{dt}\right)$
W	$w = \frac{1}{2} L i^2$	$w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$.
Series		
Parallel.		

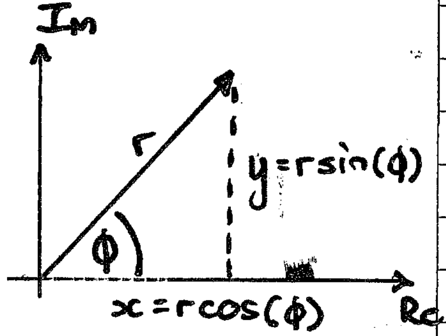
Chapter 6:

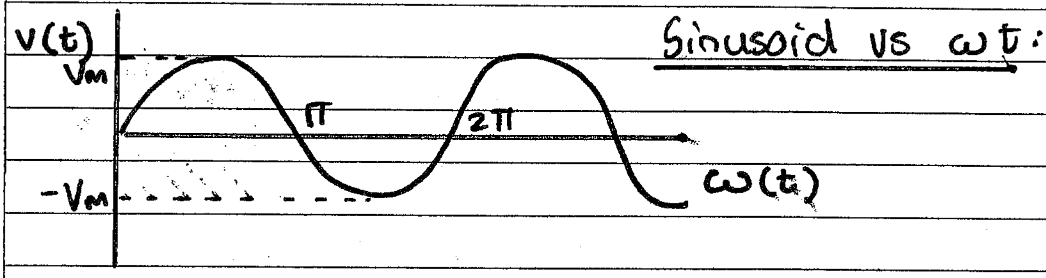
Variable	Resistor (R —WW—)	Capacitor (C —||—)	Inductor. (L —mmm—)
V_{across}	$V = i \times R$	$V = \frac{1}{C}\int i\,dt$	$V = L \times \frac{di}{dt}$
i	$i = \frac{V}{R}$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int v\,dt$
P (power)	$P = i^2R$ $= i \cdot V$	0	0
W (energy)	0	$E = \frac{1}{2}CV^2$	$E = \frac{1}{2}Li^2$
Series eq.	$R_T = R_1 + R_2 + \cdots$	$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots$	$L_T = L_1 + L_2 + \cdots$
Parallel eq.	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots$	$C_T = C_1 + C_2 + \cdots$	$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots$

Chapter 9: Sinusoids:

AC - alternating current.

* Signal with sin or cos function.





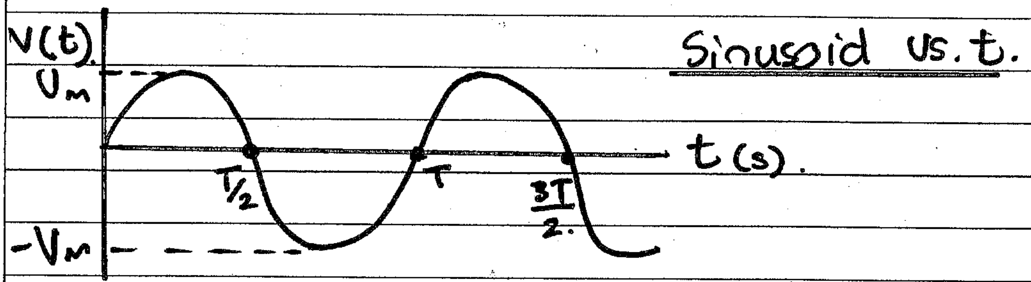
$$v(t) = V_m \sin(\omega t + \phi)$$

ϕ = phase $= \tan^{-1}\left(\frac{y}{x}\right)$

ω = angular frequency (rad/s)
V_m = amplitude of sinusoid
t = time in seconds (s)
ωt = argument of a sinusoid (rad)
ϕ = phase of sinusoid, rad or degrees

eg. $z = 2 + j6$
$r = 6.32$
$\phi = 71.57$

$z_1 = 6.32 \angle 71.56^\circ$
$= 6.32\, e^{j71.56^\circ}$



eg. $V = 3.32 \angle 32^\circ$
$x = 3.32\cos(32)$
$y = 3.32\sin(32)$. *
$V = 2.82 + j1.76$. *

T = Period

At $t = T$: $\omega t = 2\pi$ $\quad \omega T = 2\pi$
$\Rightarrow T = \frac{2\pi}{\omega}$ [s]

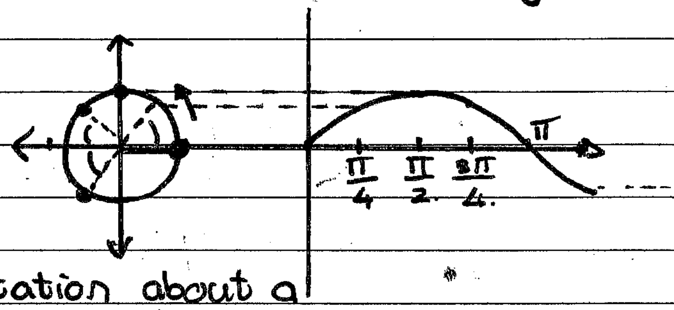
* $f = \frac{1}{T} = \frac{\omega}{2\pi}$ [Hz].

* $\omega = 2\pi f$ [rad/s]

How sinusoids came about:

$$e^{j\omega t} = \cos\omega t + j\sin\omega t.$$

ωt → rad
↳ deg.



rotation about a point in a circle.

deg → rad $\quad \varphi = \frac{\theta^\circ}{180} \times \pi$

rad → deg

Note: Let Z_1 & Z_2 = complex nrs.

$\left.\begin{array}{l} Z_1 + Z_2 \\ Z_1 - Z_2 \end{array}\right\}$ use rectangular format.

$\left.\begin{array}{l} Z_1 \times Z_2 \\ \frac{Z_1}{Z_2} \end{array}\right\}$ use polar format.

* $(r\angle\phi)^n$

$= r^n \angle(\phi \times n)$

① $a + jb$
↳ rectangular form.

② $r\angle\phi$
↳ phasor form
↳ polar form.

Z^* = conjugate.

③ $e^{j\phi}$
↳ exponential form.

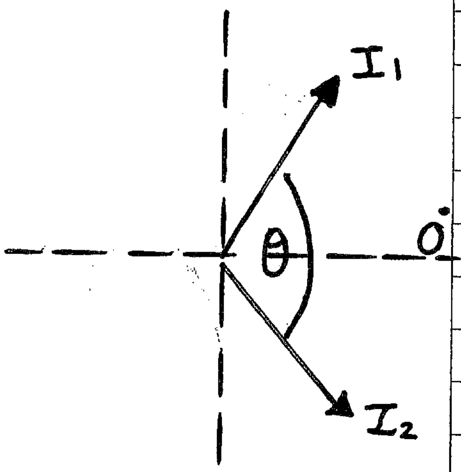
Note: $a\angle\phi = ae^{j\phi}$

EBN 122.

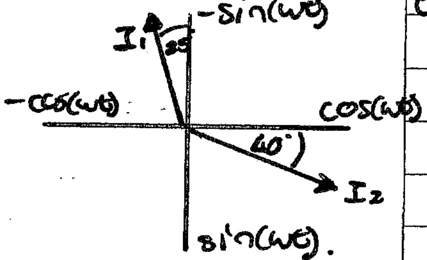
27-09-2017.

Sinusoid : Lead & Lag

- Convert to cos & < 180°
- Positive if ↺
- Which passes 0° first ?
- Both vectors start @ bottom 2 quads.



e.g. $i_1 = -4\sin(377t + 25°)$.
$i_2 = 5\cos(377t - 40°)$.



① Transform to $\cos(\omega t + \theta)$.

$i_1 = -4\cos(377t + 115°)$
$i_2 = 5\cos(377t - 40°)$

(Rotate clockwise 120°)

② Rotate i_1, i_2 to bottom Quads.

$i_1 = -4\cos(377t - 5°)$
$i_2 = 5\cos(377t - 160°)$.

$(115 + 40) = 155$

i_1 leads i_2 by 155°

③ Rotate counter clockwise, the one passes 0° first leads.

$$\Delta\phi = \phi_1 - \phi_2$$

$\Delta\phi > 0$ i_1 leads i_2.
$\Delta\phi < 0$ i_2 leads i_1.

Note: All sinusoids MUST be represented as $f(t) = A_m\cos(\omega t \pm \phi°)$.

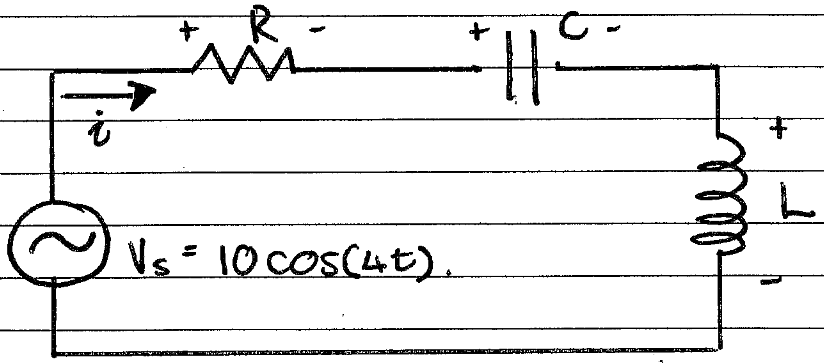
Phasor:

Analyse Sinusoid as Cosine.

deg -> rad: $\frac{\theta \pi}{180}$.

rad -> deg: $\frac{\phi\, 180}{\pi}$

Phasor & circuit Elements:



Phasor: → complex number.
→ represents amplitude & phase of a sinusoid (Cosine)

AC signal & phasor:

Euler's identity $e^{j\phi} = \cos\phi + j\sin\phi$

Understand Cosine function:

$V(t) = V_m \cos(\omega t + \phi)$

$e^{j\omega t} = \cos\omega t + j\sin\omega t$

$e^{j(\omega t+\phi)} = \cos(\omega t+\phi) + j\sin(\omega t+\phi)$.

$\Rightarrow V(t) = V_m\cos(\omega t+\phi) = Re\left(V_m e^{j(\omega t+\phi)}\right)$

$= Re\left(V_m e^{j\phi}\right) \cdot e^{j\omega t}$

$= Re\left(\bar{V} \cdot e^{j\omega t}\right)$.

rotating vector.

phasor.

$\bar{V} = V_m \cdot e^{j\phi} = V_m \angle \phi$

$= x + jy$

Trig identitities: (Sinusoid Transform).

$\sin(a+b) = \sin a \cos b + \cos a \sin b.$

$\cos(a+b) = \cos a \cos b - \sin a \sin b.$

e.g. $\sin(\omega t + 180) = \sin(\omega t)\cos(180) + \cos(\omega t)\sin(180)$
$= -\sin(\omega t)$

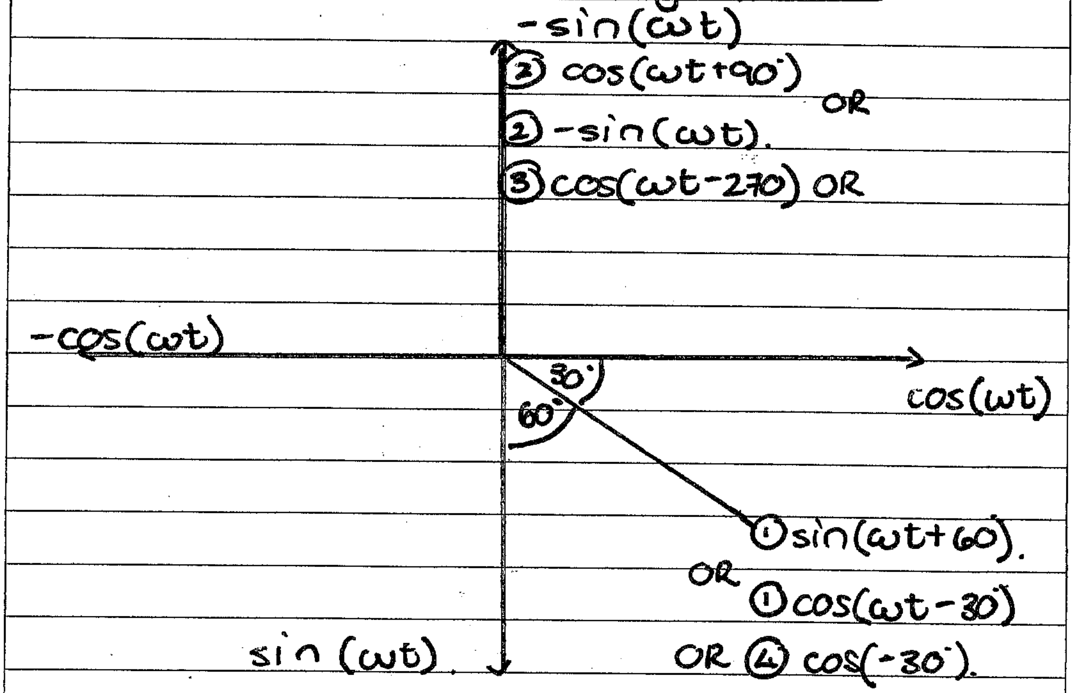
$\cos(\omega t + 180) = \cos(\omega t)\cos(180) - \sin(\omega t)\sin(180)$
$= -\cos(\omega t)$

$*\ \sin(\omega t) = \cos(\omega t - 90)$

$\sin(\omega t + 90) = \sin(\omega t)\cos(90) + \cos(\omega t)\sin(90)$.
$= \cos(\omega t)$.

$\cos(\omega t + 90) = -\sin(\omega t)$.

Sinusoid Transform (graphical).

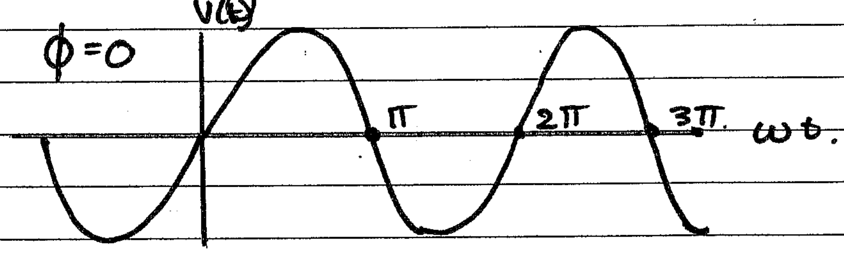


$-\cos(\omega t + 150)$
$= -\cos(\omega t + 180 - 30)$

① $\sin(\omega t + 60^\circ)$ ③ $\cos(\omega t - 270^\circ)$.
② $\cos(\omega t + 90^\circ)$. ④ $-\cos(\omega t + 150^\circ)$.

Phase angle:

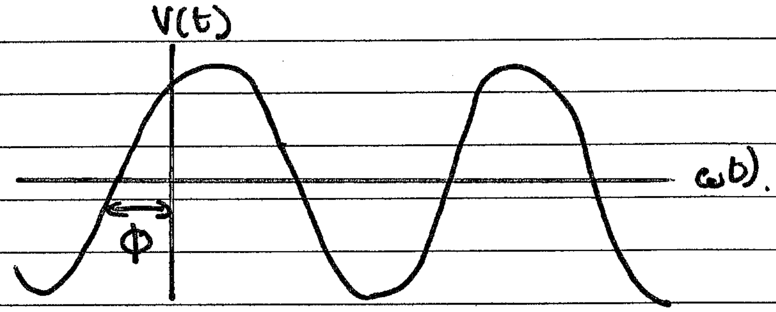
$\phi = 0$



$V(t) = V_m \sin(\omega t + \phi)$

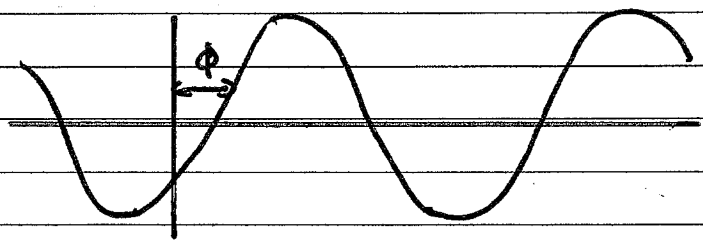
when $v(t) = (0)$ $\quad \omega t + \phi = 0$

$\omega t = 0 \Rightarrow t = 0.$



$\omega t + \phi = 0 \qquad \omega t = -\phi < 0$

$t < 0.$



$\omega t + \phi = 0. \qquad \omega t =$

e.g. $\sin(\omega t + \pi/3) \leftarrow$

$\sin(\omega t - \pi/3) \longrightarrow$

Note:
$j = \sqrt{-1}$
$= \cos(\omega t + 90)$
$= 1\angle 90^\circ$

Phasor Transformation:

$$v(t) = V_m \cos(\omega t + \phi) \leftrightarrow \bar{V} = V_m \angle \phi$$

(time domain) (phasor domain)

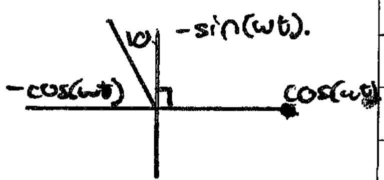
1. V_m : positive.
2. V_m : must be cosine function.
3. $-180^\circ \leq \phi \leq 180^\circ$

Time Domain Represent:		Phasor Domain rep.
$V = V_m \cos(\omega t + \phi)$	$\leftrightarrow$	$V_m \angle \phi$
$V = V_m \sin(\omega t + \phi)$	$\leftrightarrow$	$V_m \angle \phi - 90$
$i = I_m \cos(\omega t + \phi)$	$\leftrightarrow$	$I_m \angle \phi$
$i = I_m \sin(\omega t + \phi)$	$\leftrightarrow$	$I_m \angle \phi - 90$

e.g. express as a phasor & get sum.

$i_1 = 7\cos(10t + 40^\circ)$ A.
$i_2 = -4\sin(10t + 10^\circ)$ A

$i_2 = -4\cos(10t+10)$ A.



$i_2 = 4\cos(10t + 100)$ A.

$\bar{I}_1 = 7\angle 40^\circ$ A.
$\bar{I}_2 = 4\angle 100^\circ$ A

$$\bar{I}_1 + \bar{I}_2 = 7\angle 40^\circ + 4\angle 100^\circ$$
$$= \sqrt{93}\angle 61^\circ \text{ A}$$
$$= 4.67 + i\,8.44$$

Time Domain

* $\int v\,dt \longleftrightarrow \dfrac{\bar{V}}{j\omega}$

* $\dfrac{dv}{dt} \longleftrightarrow j\omega \bar{V}$

Phasor Domain.

16-10-2017.

Euler's identity:

* $Ae^{j\phi} = A\left[\cos\phi + j\sin\phi\right]$

AC signal as a Phasor:

* $v(t) = V_m \cos(\omega t + \phi)$

$= \mathrm{Re}\left\{V_m e^{j(\omega t + \phi)}\right\}$

$= \mathrm{Re}\left\{V_m e^{j\omega t} \cdot e^{j\phi}\right\}$

Let $V_m e^{j\phi} = V$

$\Rightarrow v(t) = \mathrm{Re}\left\{V e^{j\omega t}\right\}$

$\Rightarrow V = \bar{V} = V_m e^{j\phi} = V_m \angle \phi$

Transformation:

$v(t) = V_m \cos(\omega t + \phi) \longleftrightarrow \bar{V} = V_m \angle \phi$

(time domain) (phasor domain)

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16-10-2017.

Revision:

- $e^{j\phi} = \bar{V}$
- $e^{j(\omega t+\phi)}$
  $= e^{j\phi} \cdot e^{j\omega t}$
- $\omega = 2\pi f$

$$V(t) = V_m \cos(\omega t + \phi)$$

$$= Re\left[V_m \cdot e^{j(\omega t+\phi)}\right]$$

$$= Re\left[V_m \cdot e^{j\phi} \cdot e^{j\omega t}\right].$$

rotating vector. $(\bar{V})$

* $\frac{d\,(V(t))}{dt} = Re\left[j\omega \bar{V} e^{j\omega t}\right].$

↳ $\frac{d\,(\bar{V})}{dt} \longrightarrow j\omega \bar{V}$

* $\int V(t)\,dt = Re\left[\int \bar{V} e^{j\omega t} dt\right.$

$$= Re\left[\frac{\bar{V}}{j\omega} e^{j\omega t}\right].$$

↳ $\int \bar{V}\,dt \longrightarrow \frac{\bar{V}}{j\omega}$

Impedance: (Z) $[\Omega]$

* ratio between → phasor voltage. → phasor current.

$\bar{V}$ = phasor voltage.

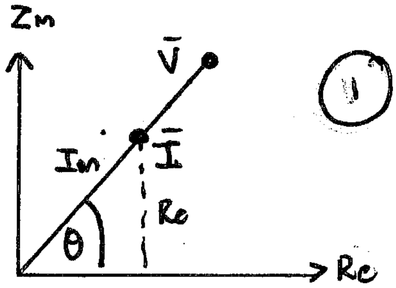
$\bar{I}$ = phasor current.

$$Z = \frac{\bar{V}}{\bar{I}} = \underbrace{R}_{\text{Resistance.}} + j\underbrace{X}_{\text{Reactance.}}$$

Admittance (Y):

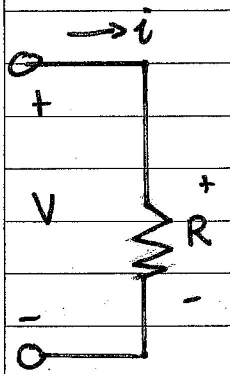
$$Y = \frac{1}{Z} = G + jB$$

Circuit Elements: (Impedence) [Z]



"In phase" = same ϕ angle.

(1)

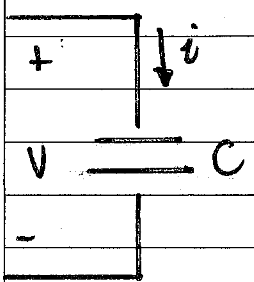


$$i(t) = I_m \cos(\omega t + \phi)$$
$$* \ \bar{I} = I_m \angle \phi$$

$$v(t) = R \cdot I_m \cos(\omega t + \phi)$$
$$* \ \bar{V} = I_m R \angle \phi$$

$$* \ Z_R = \frac{\bar{V}}{\bar{I}} = R\angle 0^\circ \Big\} \text{in phase.}$$

(2)



$$v(t) = V_m \cos(\omega t + \phi)$$
$$\bar{V} = V_m \angle \phi$$

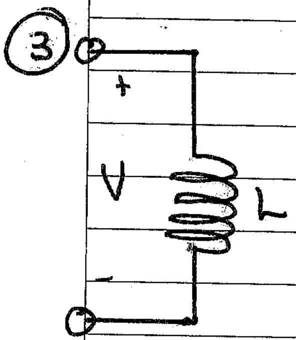
$$i(t) = C\left(\frac{dv}{dt}\right)$$
$$= C \times V_m[-\sin(\omega t + \phi)] \cdot \omega$$
$$= -\omega\left(V_m \sin(\omega t + \phi)\right) \times C$$

$$i(t) = \omega C V_m \cos(\omega t + \phi + 90^\circ)$$

$$\bar{I} = \omega\left(V_m \angle(\phi + 90^\circ)\right)$$
$$= j\omega(V_m \angle \phi)$$

$$* \ Z_C = \frac{\bar{V}}{\bar{I}} = \frac{1}{j\omega C}$$

③



$$i(t) = I_m \cos(\omega t + \phi)$$

$$\bar{I} = I_m \angle \phi$$

$$\bar{V}(t) = L \frac{d\,i(t)}{dt}$$

$$= \omega L I_m \angle \phi + 90^\circ$$

$$\bar{V} = j\omega L I_m \angle \phi$$

$$* \quad Z_L = \frac{\bar{V}}{\bar{I}} = j\omega L.$$

Inductor:

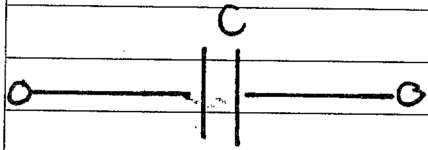
$Z_L = j\omega L \quad (\Omega)$

$\longrightarrow \omega = 2\pi f$

(Becomes short circuit in DC)

(Becomes open circuit in AC).

Capacitor:

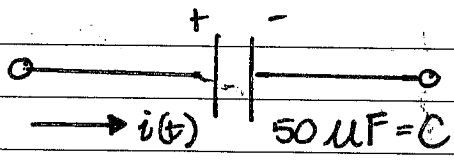


$$Z_c = \frac{1}{j\omega C} = -j\frac{1}{\omega C} \quad (\Omega)$$

(Becomes open circuit in DC)

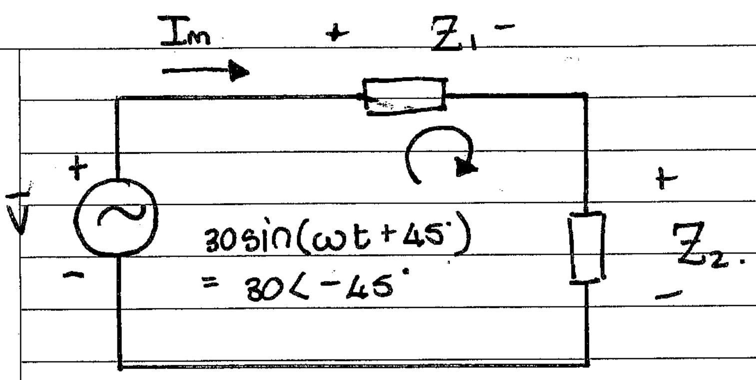
(Becomes short circuit in AC)

e.g. $V = 10\cos(100t + 30^\circ)$ $\qquad \bar{V} = 10\angle 30^\circ$



$$Z_c = \frac{1}{j(100)(50\times10^{-6})}$$

$$\bar{I}_c = \frac{\bar{V}}{Z_c}$$



$Z_T = Z_1 + Z_2$

Apply KVL: $-\bar{V} + \bar{V}_{Z_1} + \bar{V}_{Z_2} = 0$

Ohm's Law: $V_{Z_1} = \bar{I} \times Z_1$

Voltage Division: $V_{Z_1} = \bar{V} \times \dfrac{Z_2}{Z_1 + Z_2}$

Summary of Voltage-Current relat.:

Element	Time Domain	Freq. Domain	Impedence	Admittance
Resistor (R)	$v(t) = R\,i(t)$	$V = R \cdot I$	$Z = R$	$Y = \dfrac{1}{R}$
Inductor (L)	$v(t) = L\dfrac{di}{dt}$	$V = j\omega L \cdot I$	$Z_L = j\omega L$	$Y_L = \dfrac{1}{j\omega C}$
Capacitor (C)	$i(t) = C\dfrac{dv}{dt}$	$V = \dfrac{I}{j\omega C}$	$Z_c = \dfrac{1}{j\omega C}$	$Y_c = j\omega C$

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Example:

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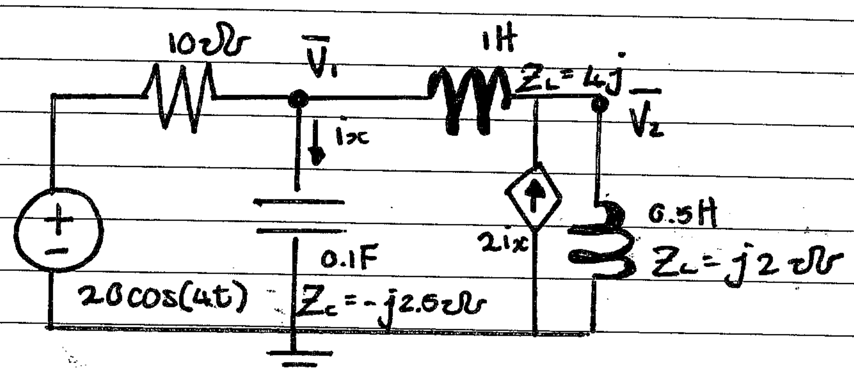


Step 1: $V(t) = 20\cos 4t$

① $V = 20\angle 0^\circ$

② $Z_r = 10\,Ω$

③ (||) $Z_c = \dfrac{1}{j\omega C} = -j\dfrac{1}{(4)(0.1)} = -j2.5\,Ω$

ω - get from source.

④ $Z_{L_1} = j\omega L = j4\,Ω$

⑤ $Z_{L0.5} = j\omega L = j2\,Ω$

Step 2:

Note: source transform would yield 1 equation for nodal.

Use Nodal Analysis : (KCL)

$\bar{V}_1$: $\dfrac{V_1 - 20\angle 0^\circ}{10\,Ω} + \dfrac{\bar{V}_1}{-j2.5} + \dfrac{\bar{V}_1 - \bar{V}_2}{j4} = 0.$ ①

$\bar{V}_2$: $\dfrac{\bar{V}_2 - \bar{V}_1}{j4} + \dfrac{\bar{V}_2}{j2} - \dfrac{2\cdot\bar{V}_1}{-j2.5} = 0$ ②

* $i_x = \dfrac{\bar{V}_1}{-j2.5}.$

① $(-3+j2)\bar{V}_1 + 5\bar{V}_2 = j40$

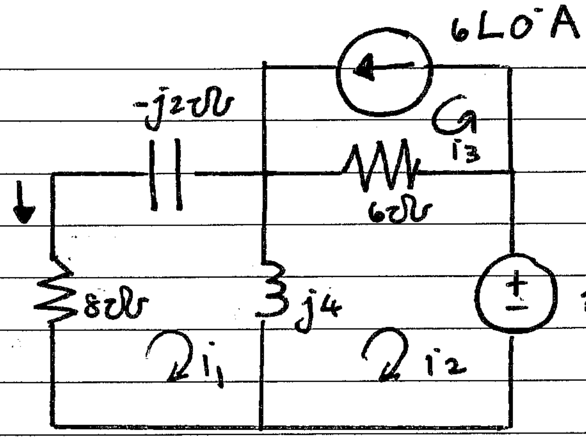
② $11\bar{V}_1 + 15\bar{V}_2 = 0$

$$\begin{bmatrix} -3+j2 & -5 \\ 11 & 15 \end{bmatrix}\begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} j40 \\ 0 \end{bmatrix}$$ - cramer's rule.

$\bar{V}_1 = 18 + j6.$

$i_x = -2.4 + 7.2j = 7.6\angle 108.4^\circ$

$i_x(t) = 7.59\cos(4t + 108.4^\circ)$



$i_3 = 6\angle 0^\circ$

$\bar{I}_1 : 8I_1 + (-j2)I_1 + j4(I_1 - I_2) = 0$

$\bar{I}_2 : j(4)(I_2 - I_1) + 6(I_2 + 6) + 30\angle 30^\circ = 0.$

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23-10-2017.

Chapter 10: Sinusoidal Circuit Theorems:

Steps of AC circuit:

1) Transform to phasor domain.

2) Solve

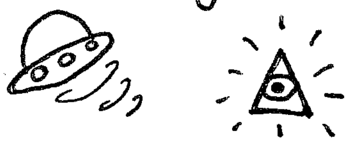
3) Transform to time domain:

Nodal Analysis: (Solve same as chap 4)

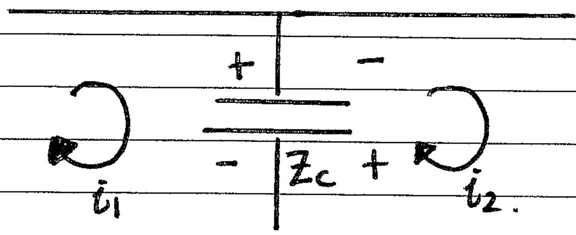
* Note: $\frac{1}{j} = -j$ & $\frac{1}{-j} = j$

Mesh Analysis:

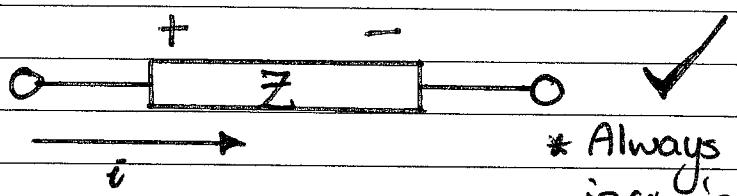
* Take note of the signs...



①



②



* Always +ve to incoming current.

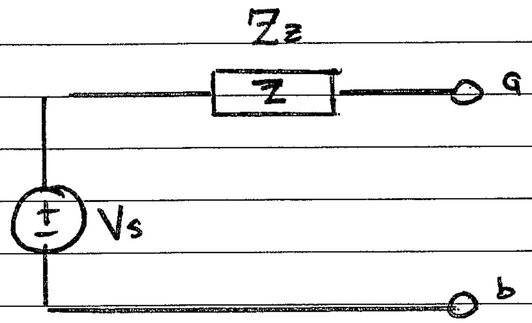
Superposition: (Same Same)

* (→) = open circuit.

* (− +) = short circuit.

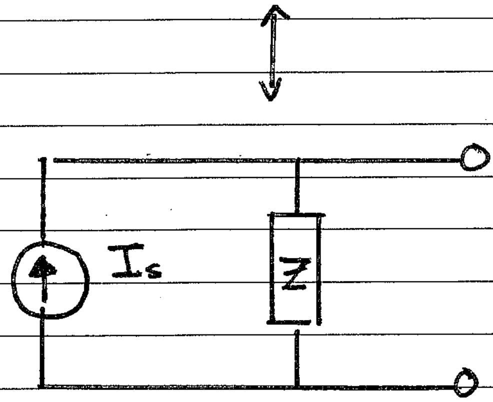
Source Transformation:

Thevenin Equivalent.



$V_s = Z_s \times I$

Norton Equivalent.



$I_s = \frac{V_s}{Z_s}$

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Circuits:

Steps:

1. Change Time domain ⟶ Phasor domain.
2. Solve complex equation.
3. Change Phasor domain ⟶ Time domain.